E-Commerce Data Analytics Project

رواد مصر الرقمية

فريق العمل

- سعيد حامد سعيد محمد
- رنا أسامة محمد فضل
- محمد شريف
- تسنيم اشرف محمد حامد
- ميار عادل شادي
- محمد الدسوقي عبده

تحت اشراف

م/عمار مصطفى

Project Documentation Guidelines

رواد مصر الرقمية

Graduate Deadline	Item
1/27/2026	Project Planning & Management
3/1/2026	Literature Review
3/1/2026	Requirements Gathering
4/3/2026	System Analysis & Design
5/17/2026	Implementation (Source Code & Execution)
5/24/2026	Final Presentation & Testing& Reports

Project Planning & Management

1

رواد مصر الرقمية

At this stage, the work was done as follows:

1. 📌 **Project Proposal** - This project aims to analyze e-commerce data and transform raw data into actionable insights to support business decision-making and improve overall performance.

2. 🎯 **Objectives** -

- Analyze customer behavior
- Measure sales performance (Revenue, Orders, AOV)
- Analyze payment methods
- Evaluate customer reviews
- Identify top products and regions

Project Planning & Management

1

رواد مصر الرقمية

3

Project Plan -

- **Track 1:** Python + SQL + Power BI dashboard
- **Track 2:** Tableau Dashboard
- **Track 3:** Excel Dashboard

✓ Implemented:

- Data Collection
- Data Cleaning
- Data Analysis
- Database Integration
- Dashboard Development
- Insights & Recommendations

Project Planning & Management

رواد مصر الرقمية

4

👥 **Roles & Task Assignment** - The project was executed through a collaborative team effort, where all members actively participated in every technical and analytical phase to ensure seamless integration and knowledge sharing:

- **Data Processing (Python)** : The team collaborated on data cleaning, handling missing values, and performing Exploratory Data Analysis (EDA) to ensure high data quality.
- **Database Management (SQL)** : All members contributed to database schema design, writing complex SQL queries, and managing the ETL pipeline to SQL Server.
- **Data Visualization (Power BI / Excel / Tableau)** : Every team member participated in designing and developing interactive dashboards across all three analytical tracks.

Project Planning & Management

5 Risk Assessment -

Risk	Mitigation
Missing Data	Data Cleaning
Duplicate Data	Removing duplicates
Data Quality Issues	Data validation

6 KPIs (Key Performance Indicators) - Project Success Metrics

- **Data Accuracy**: Achieve 100% data integrity during the ETL process from Python to SQL Server.
- Response Time: Maintain SQL query execution and dashboard loading time under 3 seconds.
- System Efficiency: 100% automation of data cleaning and transformation pipelines.
- User Adoption: Multi-platform accessibility through Power BI, Tableau, and Excel dashboards.

Literature Review

2

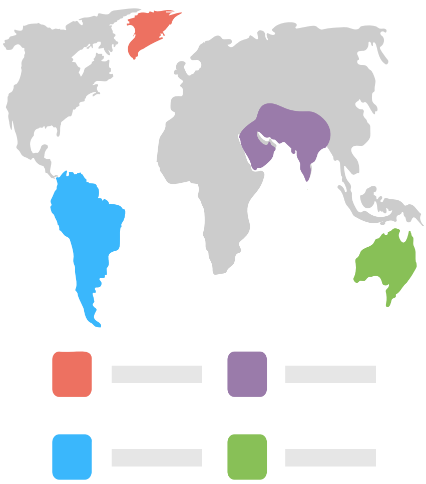
1 Importance of Data Analysis

E-commerce data analytics is essential for:

- Improving decision-making
- Increasing revenue
- Enhancing customer experience

2 Key Concepts Used

- Data Cleaning Techniques
- Exploratory Data Analysis (EDA)
- Business Intelligence Dashboards
- Sales & Customer Analysis



3 Suggested Improvements

- Machine Learning models
- Forecasting techniques
- Recommendation systems

Requirements Gathering

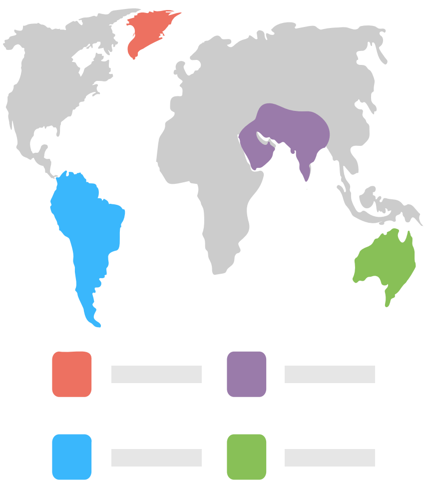
رواد مصر الرقمية

1 Stakeholders

- Management
- Marketing Team
- Operations Team

2 User Stories

- As a manager, I want to monitor total revenue
- As an analyst, I want to analyze performance trends
- As a user, I want a clear and interactive dashboard



3 Functional Requirements

- Load data from CSV files
- Clean and preprocess data and Merge datasets
- Perform data analysis
- Visualize results in dashboards

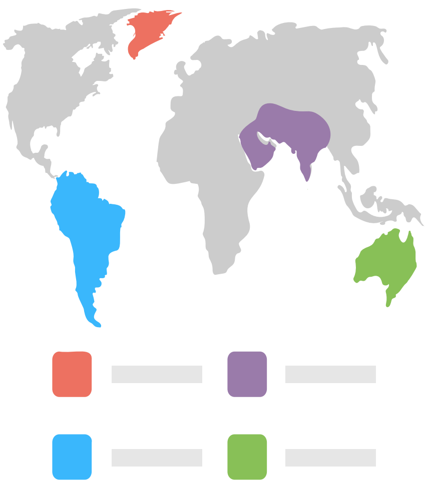
Requirements Gathering

3

4 Business & Analytical Questions: To guide the analysis and ensure alignment with business objectives, the following key analytical questions were defined:

1. Key Performance Indicators (KPIs)

- What is the Total Revenue generated?
- What is the Total number of unique Orders placed?
- What is the Average Order Value (AOV)?
- What is the Average Shipping Cost (Freight Value) per order?
- What is the Average Delivery Time in days?
- What is the Average Review Score given by customers?
- What are the Total Cancelled Orders and the overall Cancellation Rate %?



2. Time-Based Analysis

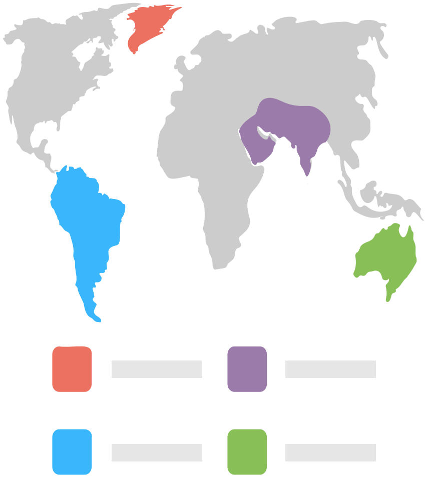
- How do Orders change over time (Years, Months)?
- What is the Monthly and Yearly Revenue Trend?

Requirements Gathering

- What time of day (Hour) do customers most frequently place orders?
- How are orders distributed by Day of the Week?
- Is there a clear Seasonal Pattern in sales performance?

3. Customer & Geographic Analysis

- Which States have the highest number of customers?
- Which Cities generate the most orders?
- Who are the Top 10 Customers based on total payment value?
- What is the ratio of Returning Customers vs. Total Customers?
- How is customer density distributed on a Geographic Concentration Map?



4. Product & Category Analysis

- Which Product Categories generate the highest revenue?
- What are the Top 5 best-selling products by sales value?
- What are the Bottom 5 lowest-selling products?

Requirements Gathering

- Is there a correlation between Product Price and Freight Cost?
- Is there a relationship between Product Weight and Freight Cost?

5. Operations & Logistics Analysis

- What is the distribution of Order Statuses (Delivered, Shipped, Canceled, etc.)?
- What is the Delivery Performance in terms of "On Time" vs. "Late" shipments?
- Which Shipping Methods are used the most and what is their impact on delivery days?
- How are Sellers distributed across different states?

6. Payments & Customer Satisfaction

- What is the revenue distribution by Payment Type (Credit Card, Boleto, Voucher, Debit Card)?
- How are Review Scores distributed (1 to 5 stars)?
- What is the Average Review Score by Order Status?
- What is the total number of Cities served by the platform?

System Analysis & Design

4

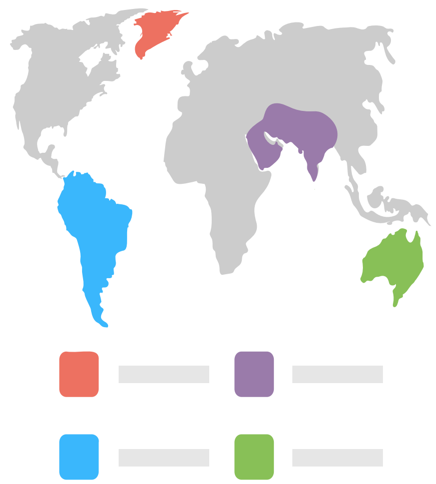
1 📌 **Problem Statement :** Raw data is unstructured and cannot be directly used for decision-making.

2 🏗 System Architecture and Data Flow

CSV Files → Python → SQL Server → Dashboards

Raw Data → Cleaning → Processing → Analysis → Visualization

1. Import Libraries
2. Reading Data
3. Explore Data
4. Data Cleaning
5. Exploratory Data Analysis (EDA)
6. create Dashboard
7. Exporting Data to SQL Database



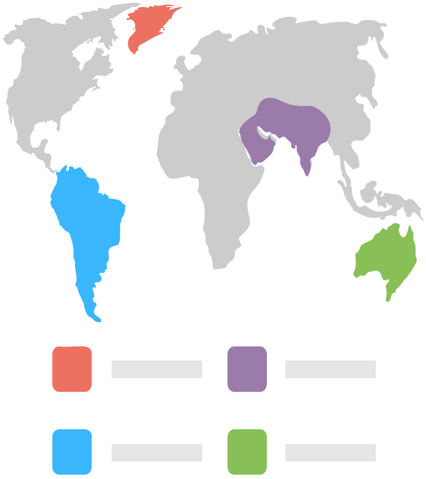
System Analysis & Design

3 Data Cleaning (Example)

- Purpose: Convert columns to Date Time format:

```
11]: date_columns=['order_purchase_timestamp','order_approved_at','order_delivered_carrier_date','order_delivered_customer_da
     for col in date_columns:
         Orders[col]=pd.to_datetime(Orders[col])
```

```
[15]: items['shipping_limit_date']=pd.to_datetime(items['shipping_limit_date'])
      items.info()
```



- Purpose: Add anew columns as a month and day name:

```
[13]: Orders['month'] = Orders['order_purchase_timestamp'].dt.to_period('M')
      Orders['day_name'] = Orders['order_purchase_timestamp'].dt.day_name()
```

3 Data Cleaning (Example):

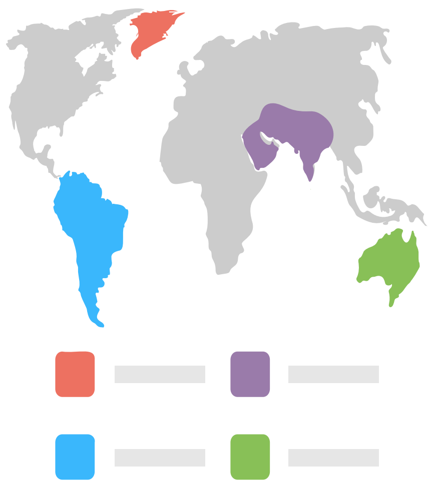
- Purpose: Handle missing values

```
[12]: Orders[date_columns]=Orders[date_columns].fillna(pd.NaT)
```

4 Data Integration:

- Purpose: Create a unified dataset

```
[19]: df = Orders.merge(customers,on='customer_id')
      df = df.merge(items,on='order_id')
      df = df.merge(products,on='product_id')
      df = df.merge(payments,on='order_id')
      df = df.merge(reviews,on='order_id', how='left')
      df.head()
```



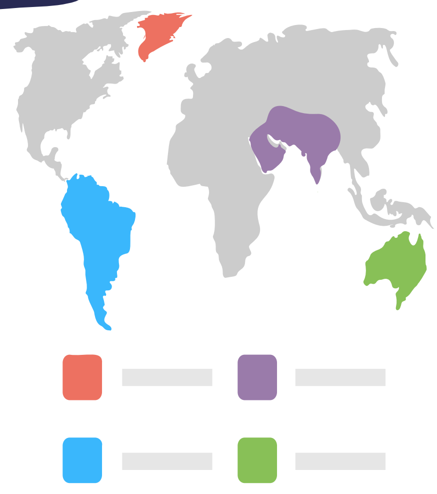
System Analysis & Design

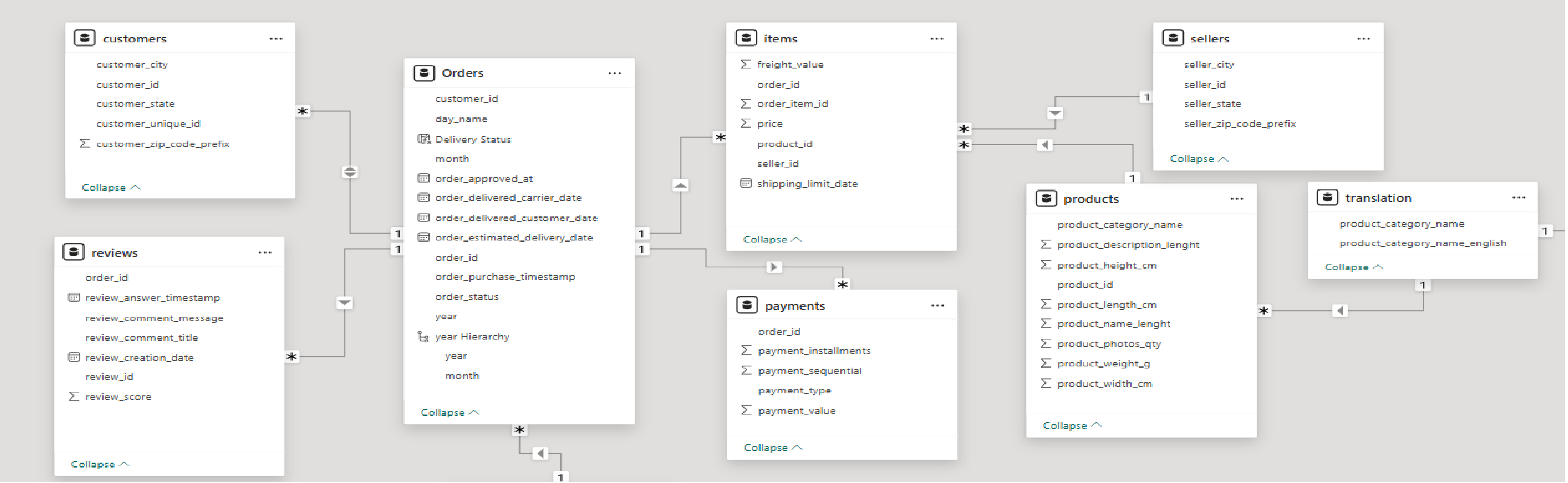
5 Database Design & Data Modeling

Customers → Orders (1:N)

Orders → Items (1:N)

Orders → Payments (1:N)





Implementation (Source Code & Execution)

5

✓ Tracks Implementation :

Track 1 : Python + SQL + Power BI - Implemented:

Data Cleaning > Data Analysis > Database Integration > Dashboard Development

2. Reading Data

```
[6]: customers= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\customers.csv")
Orders= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\Orders.csv")
items= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\order_items.csv")
products= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\products.csv")
payments= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\order_payments_dataset.csv")
sellers= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\sellers.csv")
reviews= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\order_reviews.csv")
geo= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\geolocation.csv")
translation= pd.read_csv(r"E:\مبادرة رواد مصر الرقمية\رواد 4\مشروع التخرج\product_category_name_translation.csv")
```

3.Explor Data

```
[8]: dfs = {
    "customers": customers,
    "Orders": Orders,
    "items": items,
    "products": products,
    "payments": payments,
    "sellers": sellers,
    "reviews": reviews,
    "geo": geo,
    "translation": translation
}

def explore_data(df, name="DataFrame"):
    print(f"--- Overview of {name} ---\n")
    print("Shape:", df.shape, "\n")
    print("Sample 5 rows:\n", df.sample(5), "\n")
    print("Columns:", df.columns.tolist(), "\n")
    print("Unique values per column:\n", df.nunique(), "\n")
    print("Info:\n")
    print(df.info(), "\n")
    print("Missing values per column:\n", df.isnull().sum(), "\n")
    print("Duplicate rows:", df.duplicated().sum(), "\n")
    print("-"*50, "\n")

for name, df in dfs.items():
    explore_data(df, name)
```

Implementation (Source Code & Execution)

5

1

Python Examples :

Question:

Q1. How many total orders do we have?

```
[24]: total_orders = Orders['order_id'].nunique()
      print("total orders:",total_orders)
```

```
total orders: 99441
```

Q2: What is the distribution of order status?

```
[26]: order_status = Orders['order_status'].value_counts()
      order_status
```

```
[26]: order_status
      delivered      96478
      shipped         1107
      canceled         625
      unavailable      609
      invoiced         314
      processing       301
      created            5
      approved           2
      Name: count, dtype: int64
```

Implementation (Source Code & Execution)

5

رواد مصر الرقمية

1

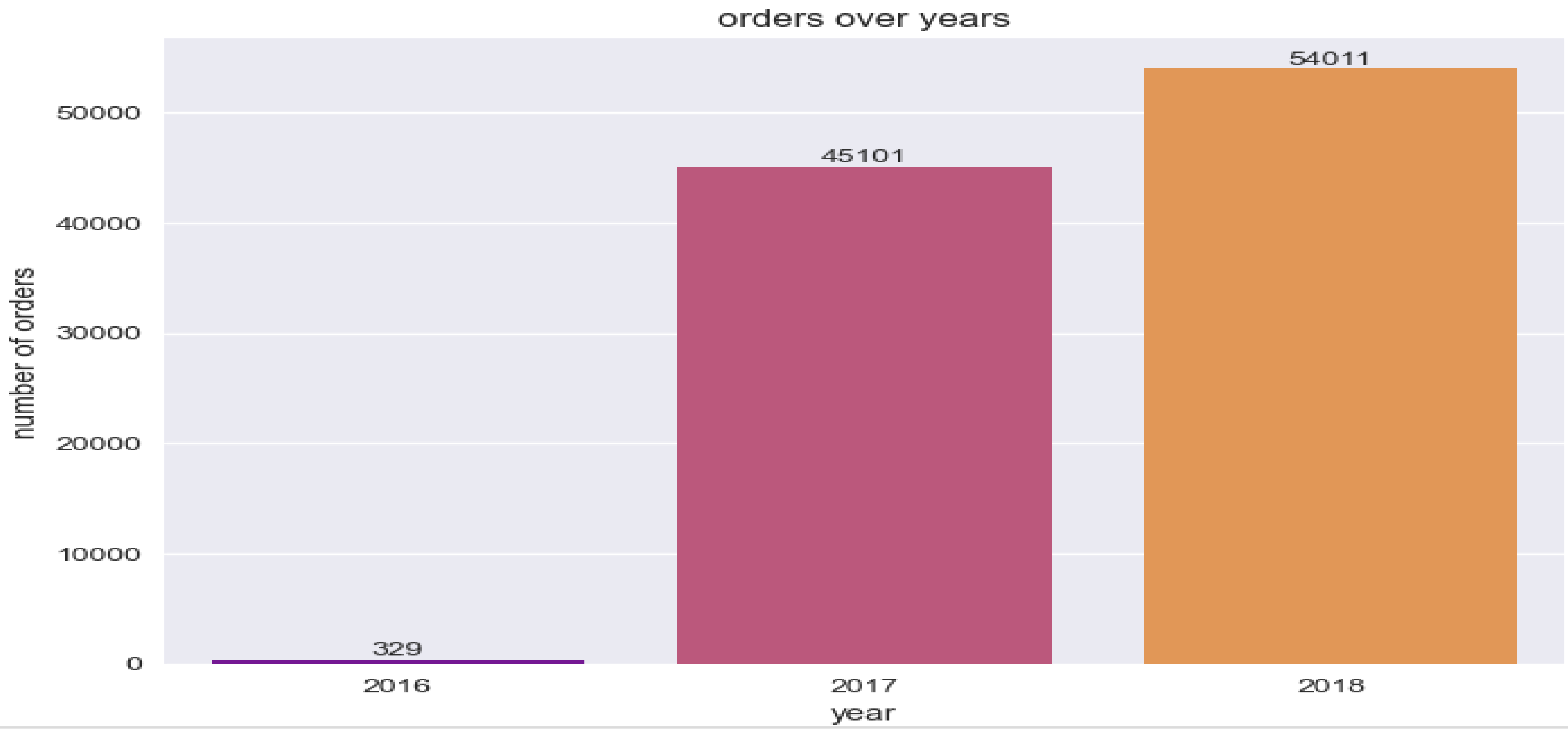
Python Examples :

Question:

Q3: How do orders change over time?

[29]:
```
Orders['year'] = Orders['order_purchase_timestamp'].dt.to_period('y')
yearly_orders = Orders.groupby('year').size()

ax=sns.barplot(x=yearly_orders.index.astype(str), y=yearly_orders.values, palette='plasma')
for i in ax.containers:
    ax.bar_label(i,)
plt.title("orders over years")
plt.xlabel("year")
plt.ylabel("number of orders")
plt.show()
```



Implementation (Source Code & Execution)

5

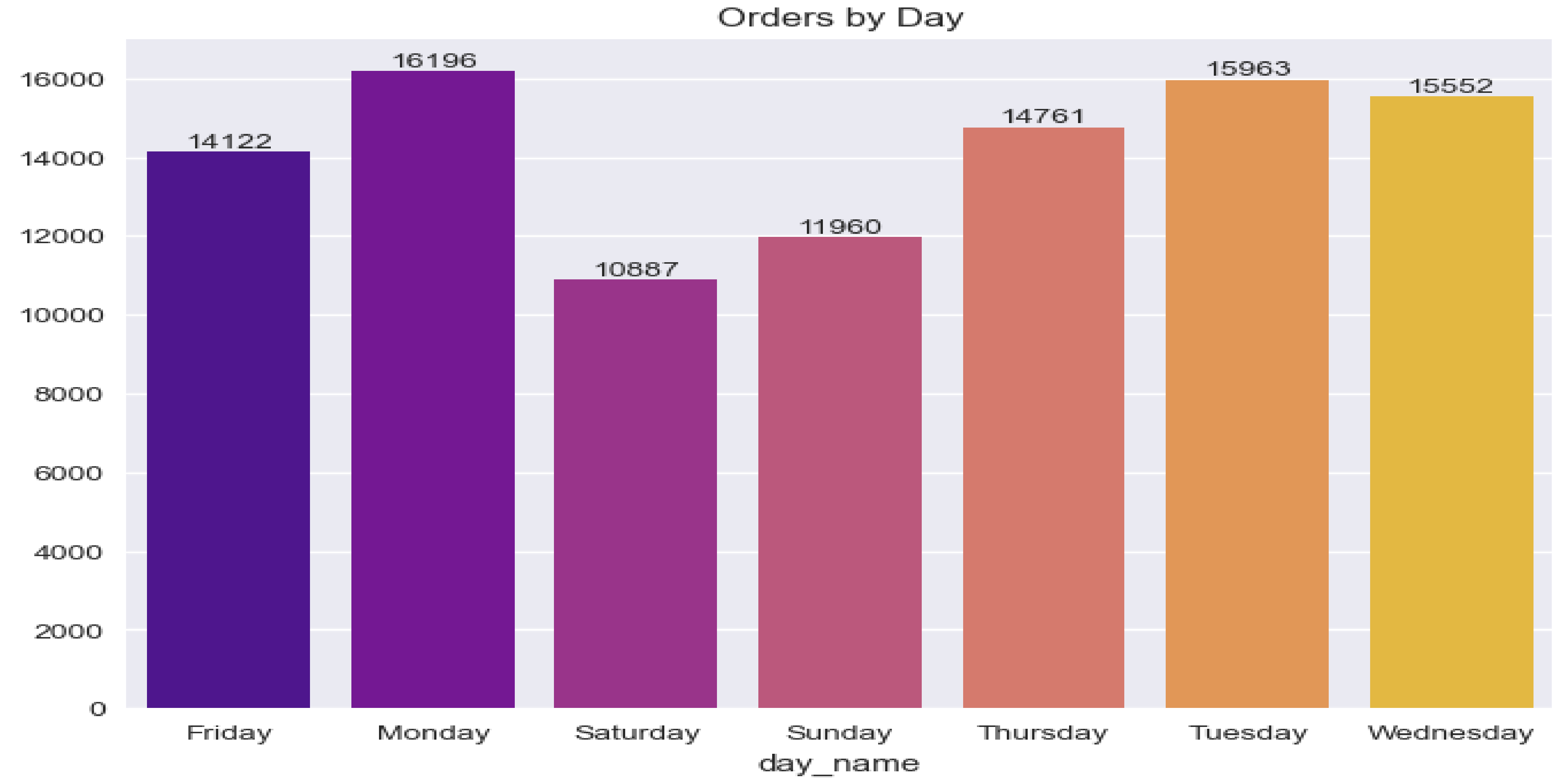
رواد مصر الرقمية

1 Python Examples :

Question:

[33]:
```
orders_day = Orders.groupby('day_name')['order_id'].count()
ax=sns.barplot(x=orders_day.index,y=orders_day.values,palette='plasma')
for i in ax.containers:
    ax.bar_label(i,)

plt.title("Orders by Day")
plt.show()
```



Implementation (Source Code & Execution)

5

رواد مصر الرقمية

1 Python Examples :

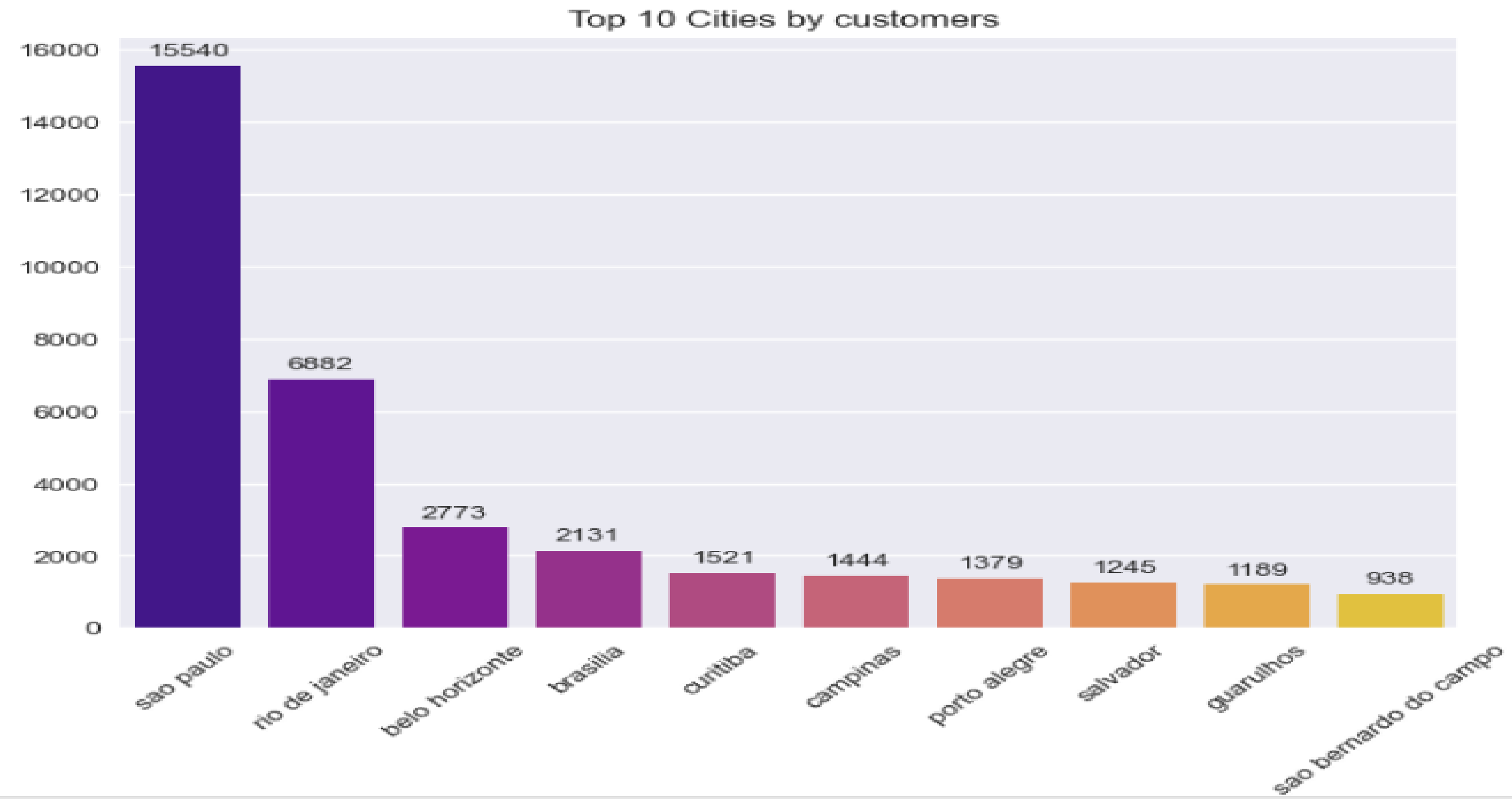
Question:

Q7: Which cities have most customers?

[45]:
```
top_cities = customers.groupby('customer_city')['customer_id'].count().sort_values(ascending=False)[:10]

ax = sns.barplot(x=top_cities.index, y=top_cities.values, palette='plasma')
plt.title("Top 10 Cities by customers ")
plt.xticks(rotation=45)

for i in ax.containers:
    ax.bar_label(i, padding=3)
plt.show()
```



Implementation (Source Code & Execution)

2 Python Dashboard :

$16,008,872

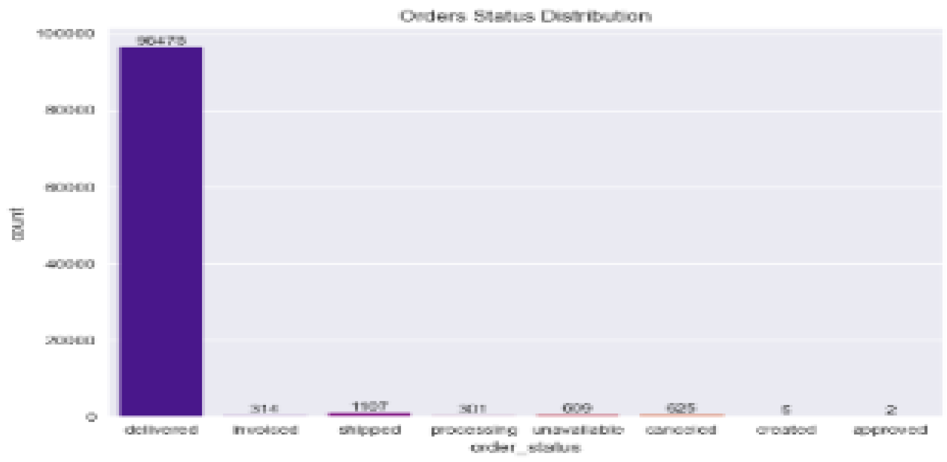
Total_Revenue

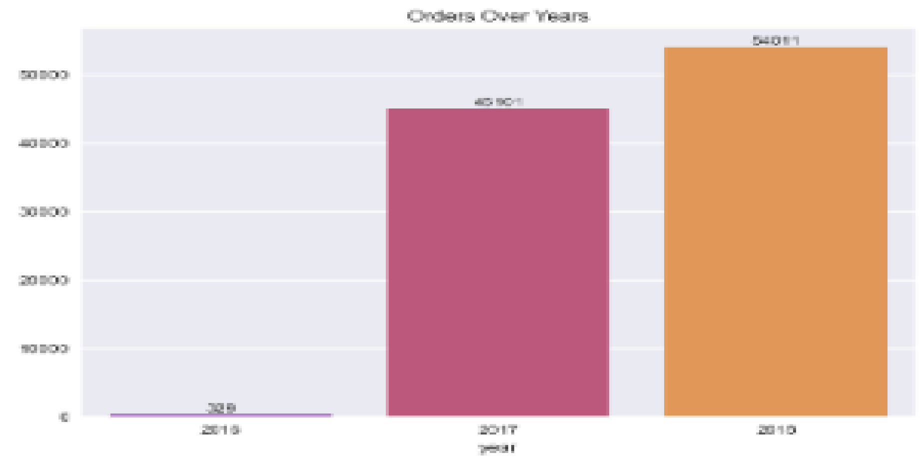
99,441

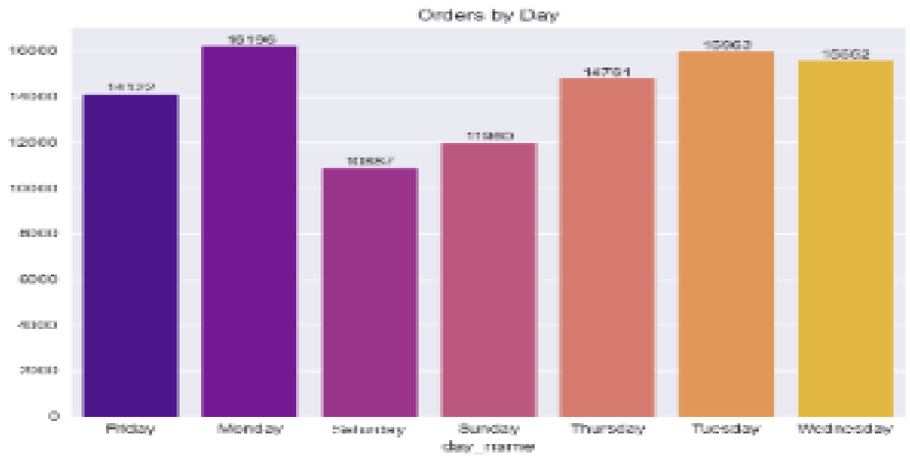
Total Orders

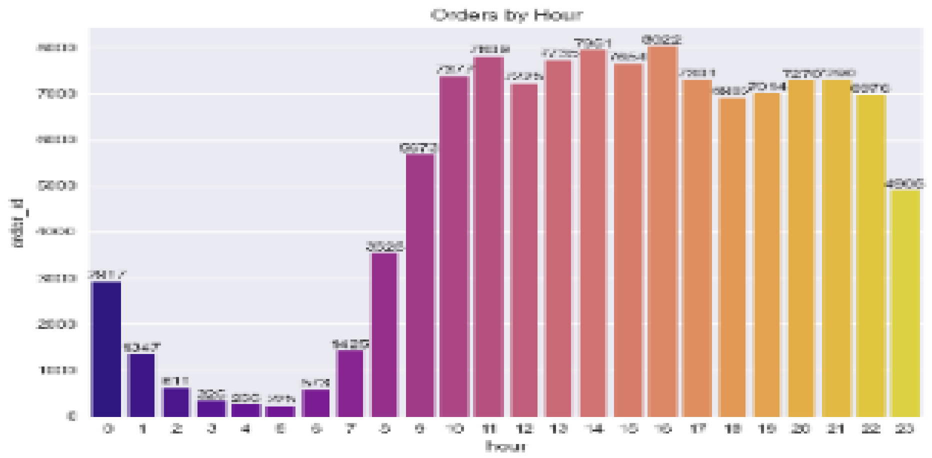
$161.00

Avg Order Value







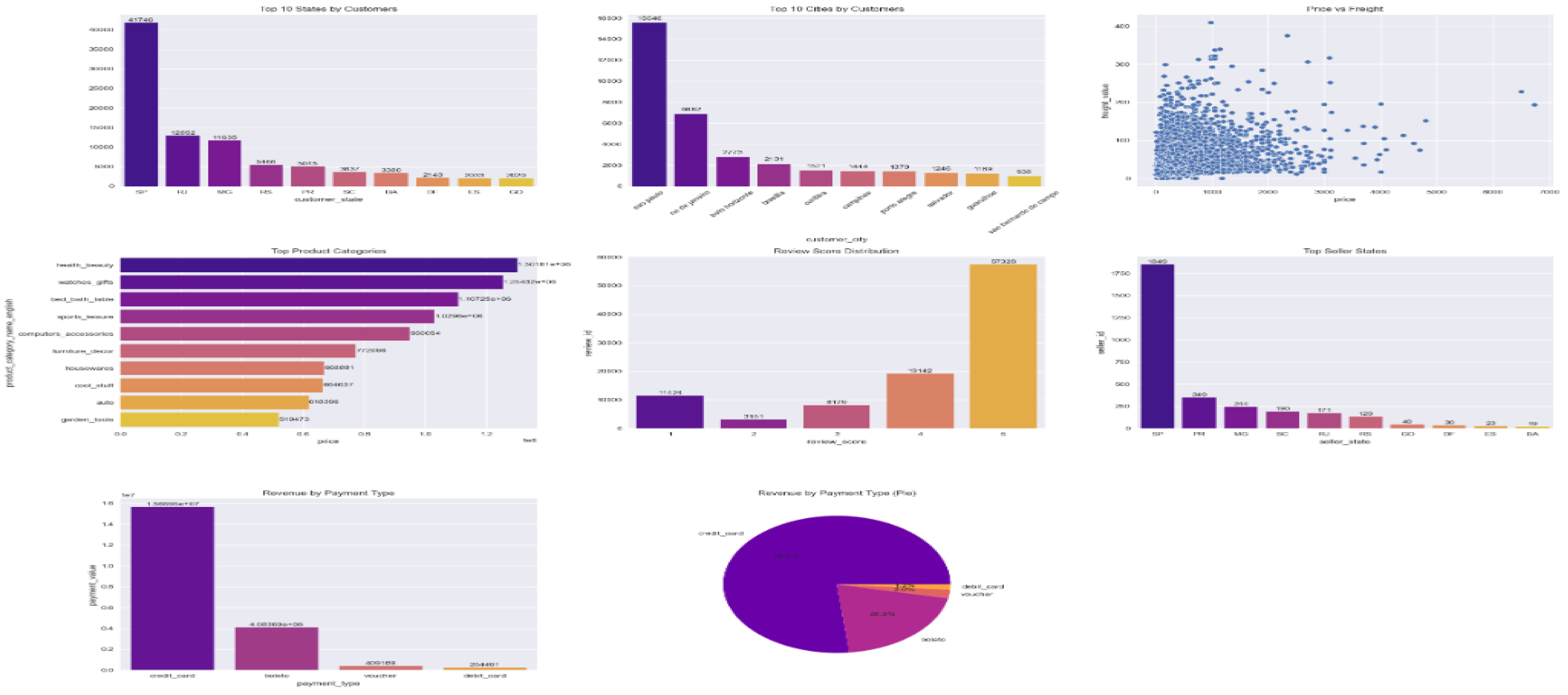


Implementation (Source Code & Execution)

5

رواد مصر الرقمية

2 Python Dashboard :



Implementation (Source Code & Execution)

5

Exporting Data to SQL Database:

```python
from sqlalchemy import create_engine


SERVER_NAME   = "BUSINESS_PLUS"
DATABASE_NAME = "E-commerce_Analytics"

connection_string = (
    f"mssql+pyodbc://{SERVER_NAME}/{DATABASE_NAME}"
    "?driver=ODBC+Driver+17+for+SQL+Server"
    "&trusted_connection=yes"
)
engine = create_engine(connection_string)
print("✅ تم الاتصال بـ SQL Server بنجاح!")


for col in Orders.columns:
    if str(Orders[col].dtype).startswith("period"):
        Orders[col] = Orders[col].astype(str)
        print(f"تم تحويل عمود '{col}' من Period إلى string")


tables = {
    "customers"   : customers,
    "Orders"      : Orders,
    "items"       : items,
    "products"    : products,
    "payments"    : payments,
    "sellers"     : sellers,
    "reviews"     : reviews,
    "geo"         : geo,
    "translation" : translation,
}

for table_name, df in tables.items():
    print(f"يرفع جدول: {table_name} ...")

    df.to_sql(
        name      = table_name,
        con       = engine,
        if_exists = "replace",
        index     = False,
    )
```

Implementation (Source Code & Execution)

5

رواد مصر الرقمية

4 SQL Examples

Question: What is the total revenue?:

Question: What is the average order value?

```
select
round(SUM(payment_value),0) as total_paymens,
round(AVG (payment_value),0) as average_order_value
from payments
```

Results | Messages

	total_paymens	average_order_value
1	16008872	154

Question: payment methods by unique order count and total value?

```
select
 payment_type,
 count( order_id)as order_count,
 round(SUM(payment_value),0) as total_paymens
 from payments
 group by payment_type
 order by  total_paymens desc
```

Results | Messages

	payment_type	order_count	total_paymens
1	credit_card	76795	12542084
2	boleto	19784	2869361
3	voucher	5775	379437
4	debit_card	1529	217990
5	not_defined	3	0

Implementation (Source Code & Execution)

5

4 SQL Examples

--average review score for orders

```sql
select
order_status,
avg(review_score) as average_score,
count(review_id) as total_reviews
from Orders o
join reviews r on o.order_id=r.order_id
group by order_status
order by average_score desc
```

Results | Messages

	order_status	average_score	total_reviews
1	delivered	4	96361
2	approved	2	2
3	shipped	2	1043
4	created	2	3
5	unavailable	1	597
6	canceled	1	609
7	invoiced	1	313
8	processing	1	296

--orders with payments higher than 4000

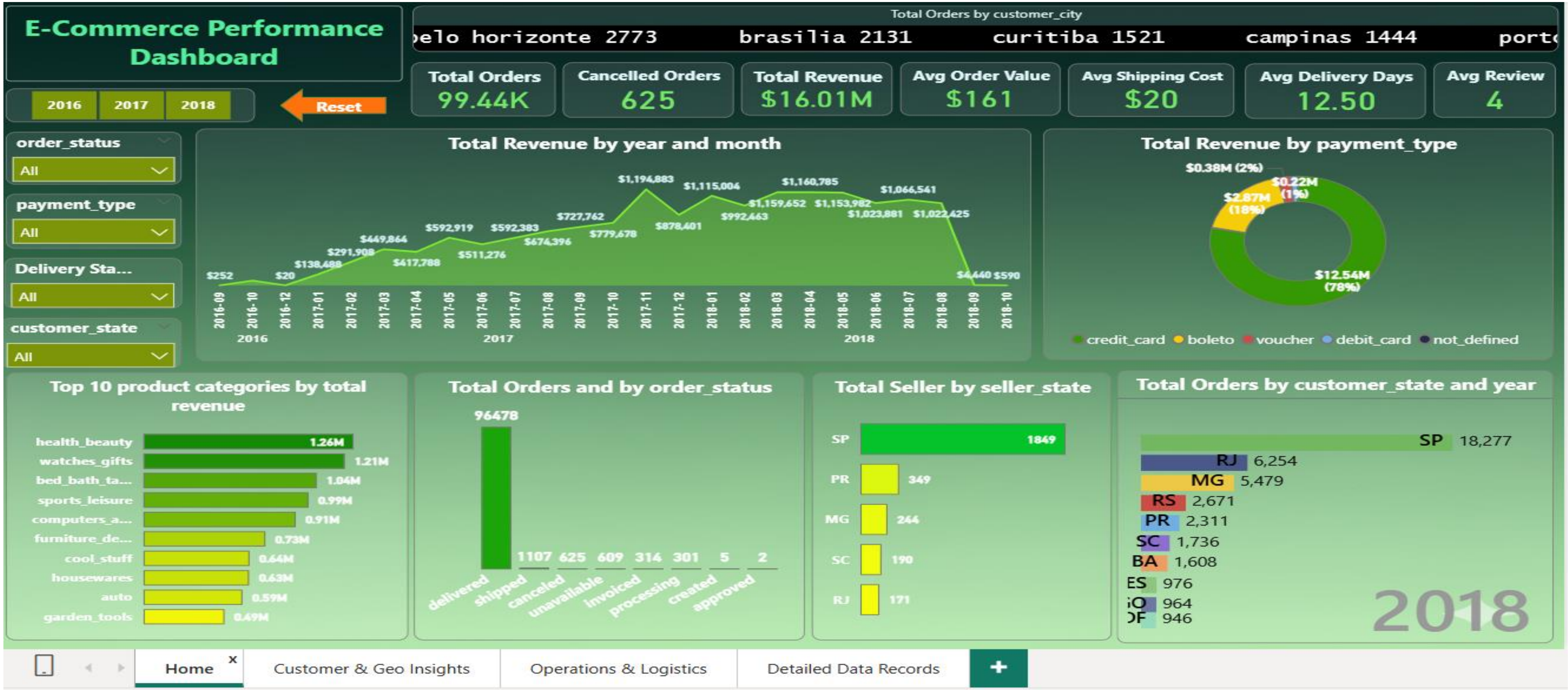
```sql
select
    order_id,
    sum(payment_value) as total_order_value
from payments
group by order_id
having sum(payment_value) > 4000
order by total_order_value desc
```

Results | Messages

	order_id	total_order_value
1	03caa2c082116e1d31e67e9ae3700499	13664.08
2	736e1922ae60d0d6a89247b851902527	7274.88
3	0812eb902a67711a1cb742b3cdaa65ae	6929.31
4	fefacc66af859508bf1a7934eab1e97f	6922.21
5	f5136e38d1a14a4dbd87dff67da82701	6726.66
6	2cc9089445046817a7539d90805e6e5a	6081.54
7	a96610ab360d42a2e5335a3998b4718a	4950.34
8	b4c4b76c642808cbe472a32b86cddc95	4809.44
9	199af31afc78c699f0dbf71fb178d4d4	4764.34
10	8dbc85d1447242f3b127dda390d56e19	4681.78
11	426a9742b533fc6fed17d1fd6d143d7e	4513.32
12	d2f270487125ddc41fd134c4003ad1d7	4445.5
13	80dfedb6d17bf23539beeef3c768f4d7	4194.76
14	68101694e5c5dc7330c91e1bbc36214f	4175.26
15	b239ca7cd485940b31882363b52e6674	4163.51
16	9a3966c23190dbdbaabed08e8429c006	4042.74

Implementation (Source Code & Execution)

5 Power BI Dashboard :



Implementation (Source Code & Execution)

5 Power BI Dashboard :

Implementation (Source Code & Execution)

5

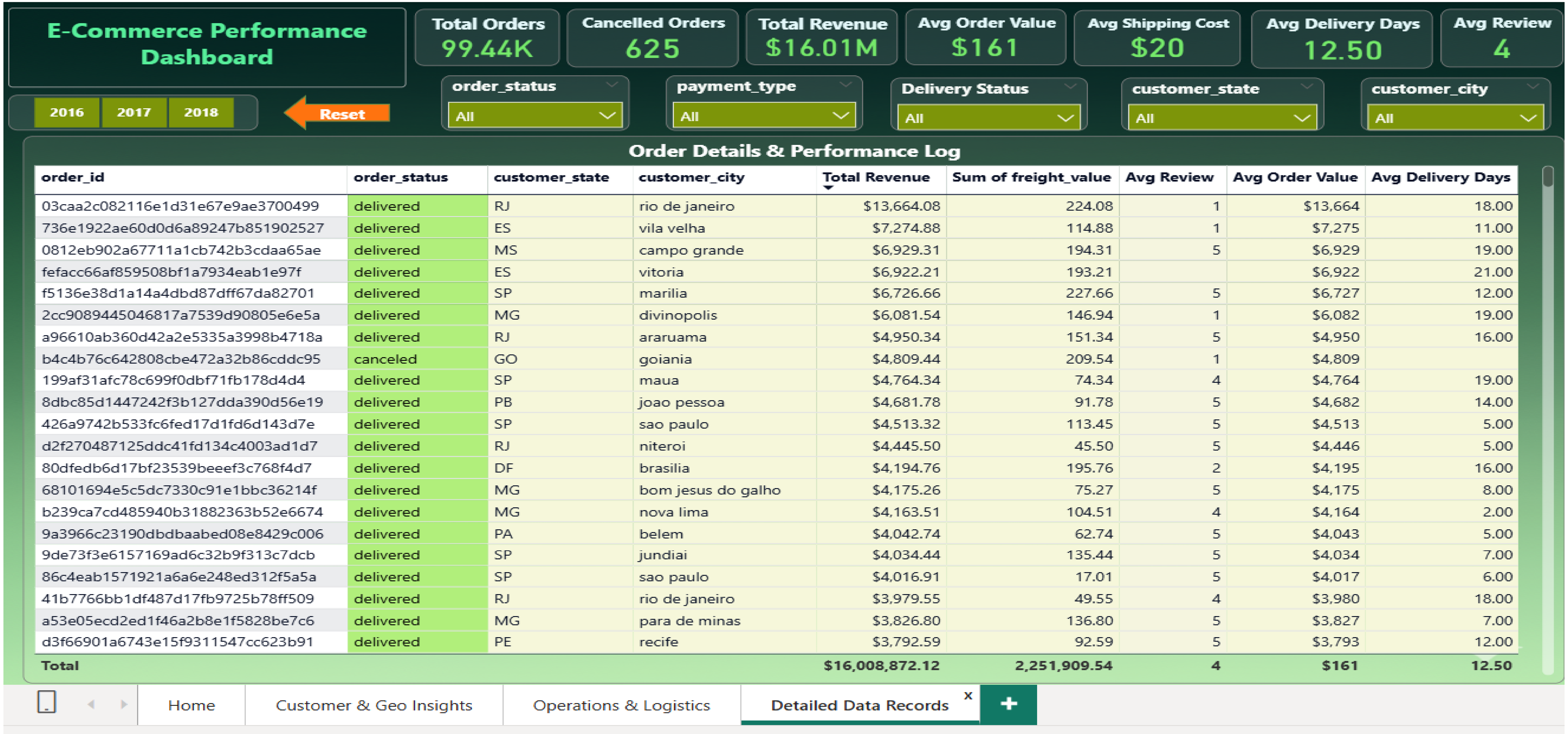
5 Power BI Dashboard :

Implementation (Source Code & Execution)

5

رواد مصر الرقمية

5 Power BI Dashboard :



order_id	order_status	customer_state	customer_city	Total Revenue	Sum of freight_value	Avg Review	Avg Order Value	Avg Delivery Days
03caa2c082116e1d31e67e9ae3700499	delivered	RJ	rio de janeiro	$13,664.08	224.08	1	$13,664	18.00
736e1922ae60d0d6a89247b851902527	delivered	ES	vila velha	$7,274.88	114.88	1	$7,275	11.00
0812eb902a67711a1cb742b3cdaa65ae	delivered	MS	campo grande	$6,929.31	194.31	5	$6,929	19.00
fefacc66af859508bf1a7934eab1e97f	delivered	ES	vitoria	$6,922.21	193.21		$6,922	21.00
f5136e38d1a14a4dbd87dff67da82701	delivered	SP	marilia	$6,726.66	227.66	5	$6,727	12.00
2cc9089445046817a7539d90805e6e5a	delivered	MG	divinopolis	$6,081.54	146.94	1	$6,082	19.00
a96610ab360d42a2e5335a3998b4718a	delivered	RJ	araruama	$4,950.34	151.34	5	$4,950	16.00
b4c4b76c642808cbe472a32b86cddc95	canceled	GO	goiania	$4,809.44	209.54	1	$4,809	
199af31afc78c699f0dbf71fb178d4d4	delivered	SP	maua	$4,764.34	74.34	4	$4,764	19.00
8dbc85d1447242f3b127dda390d56e19	delivered	PB	joao pessoa	$4,681.78	91.78	5	$4,682	14.00
426a9742b533fc6fed17d1fd6d143d7e	delivered	SP	sao paulo	$4,513.32	113.45	5	$4,513	5.00
d2f270487125ddc41fd134c4003ad1d7	delivered	RJ	niteroi	$4,445.50	45.50	5	$4,446	5.00
80dfedb6d17bf23539beeef3c768f4d7	delivered	DF	brasilia	$4,194.76	195.76	2	$4,195	16.00
68101694e5c5dc7330c91e1bbc36214f	delivered	MG	bom jesus do galho	$4,175.26	75.27	5	$4,175	8.00
b239ca7cd485940b31882363b52e6674	delivered	MG	nova lima	$4,163.51	104.51	4	$4,164	2.00
9a3966c23190dbdbaabed08e8429c006	delivered	PA	belem	$4,042.74	62.74	5	$4,043	5.00
9de73f3e6157169ad6c32b9f313c7dcb	delivered	SP	jundiai	$4,034.44	135.44	5	$4,034	7.00
86c4eab1571921a6a6e248ed312f5a5a	delivered	SP	sao paulo	$4,016.91	17.01	5	$4,017	6.00
41b7766bb1df487d17fb9725b78ff509	delivered	RJ	rio de janeiro	$3,979.55	49.55	4	$3,980	18.00
a53e05ecd2ed1f46a2b8e1f5828be7c6	delivered	MG	para de minas	$3,826.80	136.80	5	$3,827	7.00
d3f66901a6743e15f9311547cc623b91	delivered	PE	recife	$3,792.59	92.59	5	$3,793	12.00
Total				$16,008,872.12	2,251,909.54	4	$161	12.50

Implementation (Source Code & Execution)

Power BI Examples :

Question: Key Performance Indicators (KPIs)

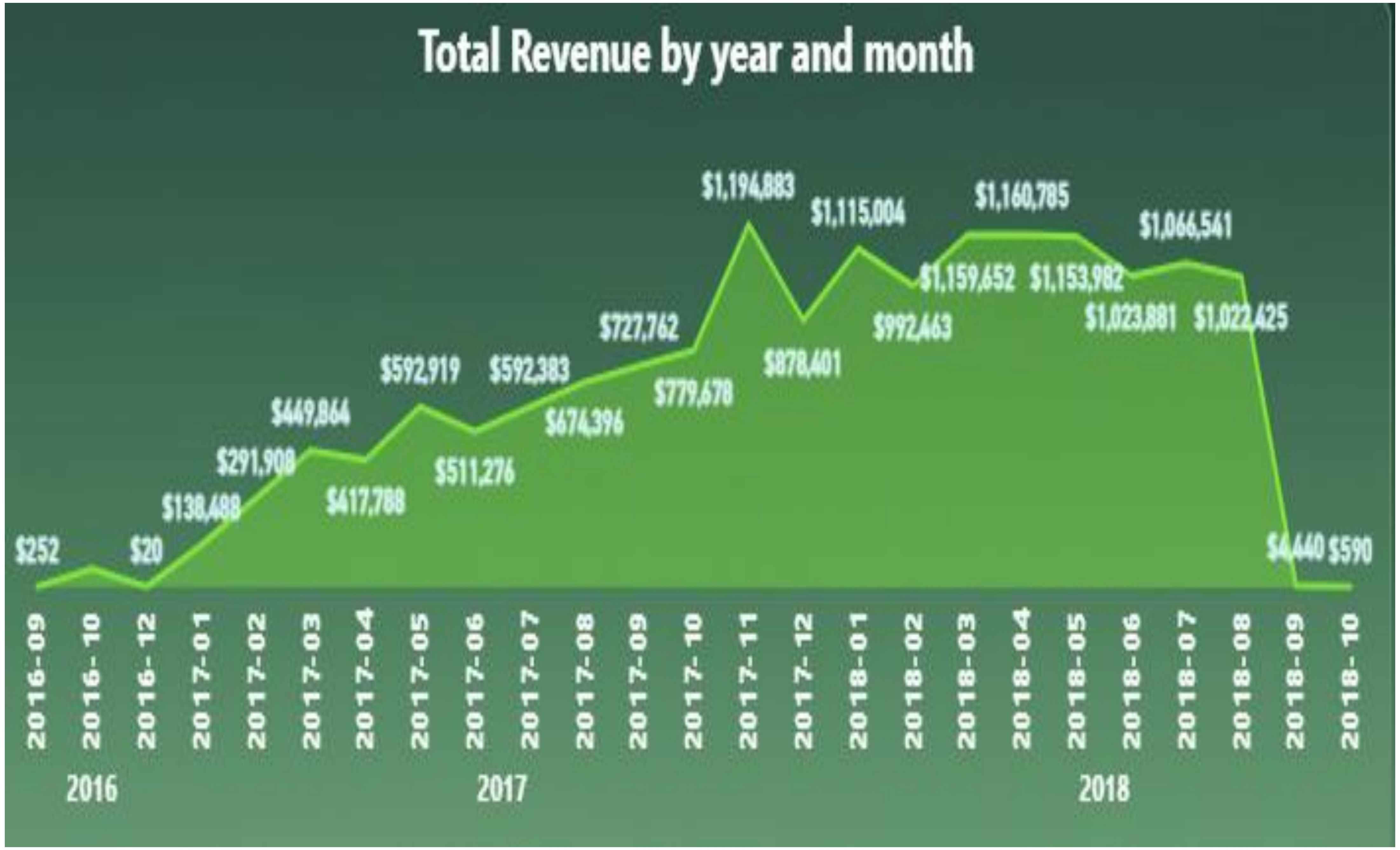
- What is the Total Revenue generated?
- What is the Total number of unique Orders placed?
- What is the Average Order Value (AOV)?
- What is the Average Shipping Cost (Freight Value) per order?
- What is the Average Delivery Time in days?
- What is the Average Review Score given by customers?
- What are the Total Cancelled Orders and the overall Cancellation Rate %?

Implementation (Source Code & Execution)

Power BI Examples :

❓ Question: What is the Monthly and Yearly Revenue Trend?

Revenue shows a strong upward trend starting from early 2017, reaching its all-time peak of $1.19M in November 2017. The business maintained a stable monthly performance above $1M throughout the first half of 2018.

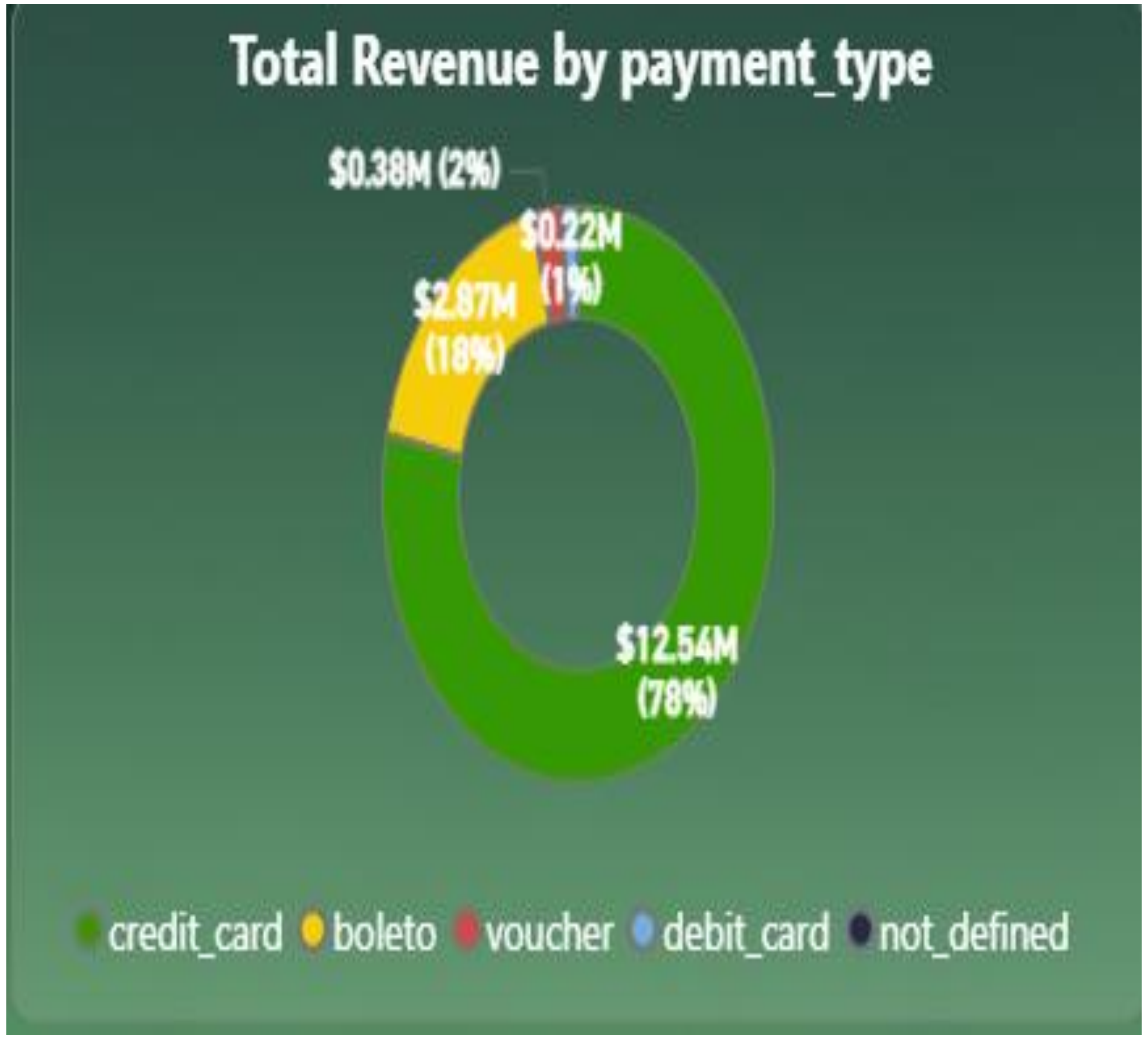


Implementation (Source Code & Execution)

Power BI Examples :

Question: • What is the revenue distribution by Payment Type?

The Credit Card is the dominant payment method, accounting for 78% ($12.54M) of total revenue, followed by Boleto at 18% ($2.87M). Other methods like Voucher and Debit Card contribute minimally, representing only 3% of the total revenue combined.



Insights :High Credit Card usage (78%) indicates a preference for installment payments, while the Boleto share (18%) emphasizes the need for cash-based options to ensure full market coverage.

Implementation (Source Code & Execution)

5

رواد مصر الرقمية

Power BI Examples :

Question: • What is the distribution of Order Statuses ?

The vast majority of orders have been successfully Delivered (96,478), while other statuses like Shipped (1,107), Canceled (625), and Unavailable (609) represent only a tiny fraction of the total volume.

Insights :The high volume of delivered orders (over 97%) indicates a very high fulfillment success rate and operational efficiency. The low number of canceled or unavailable orders suggests effective inventory management and a reliable supply chain

Implementation (Source Code & Execution)

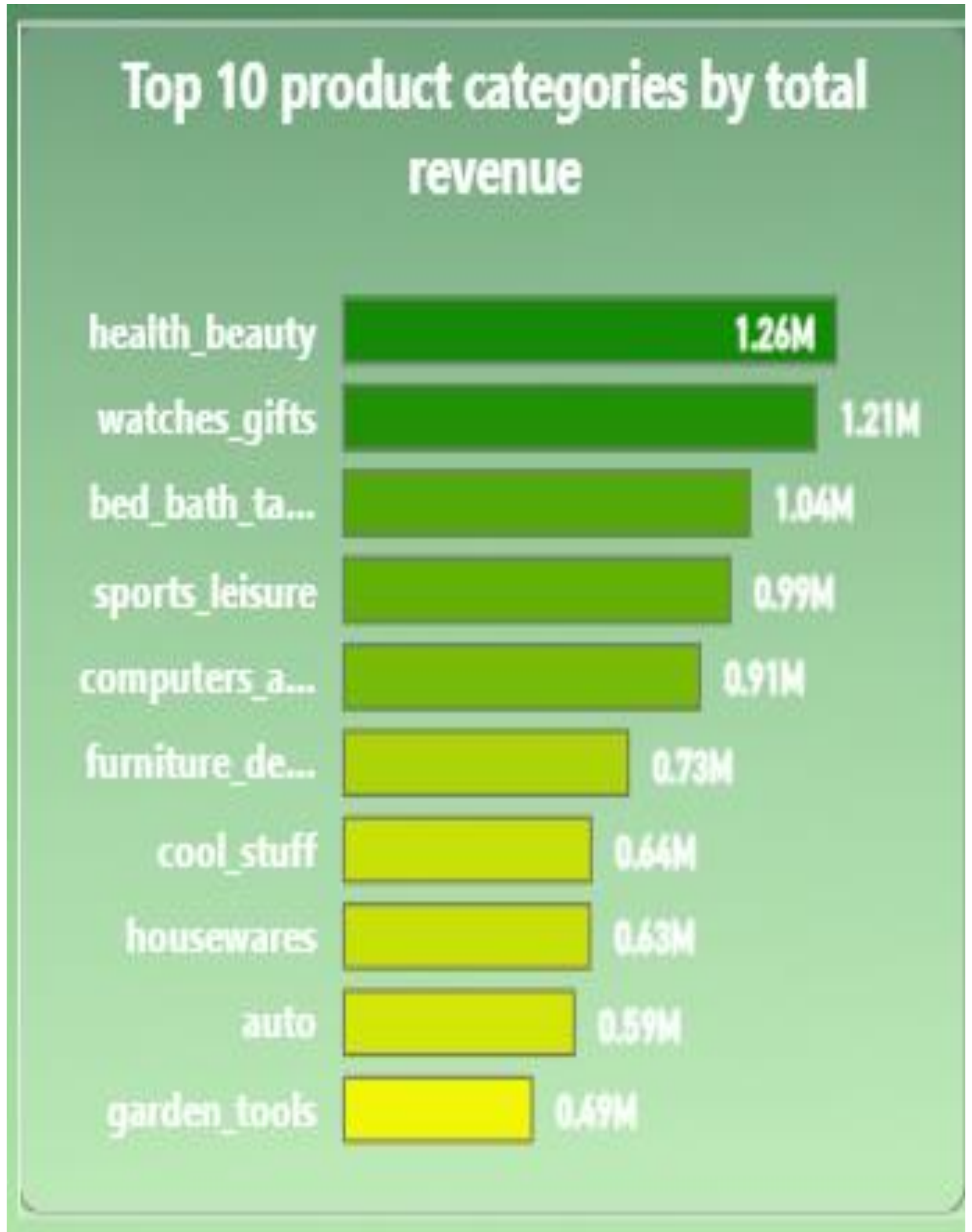
5

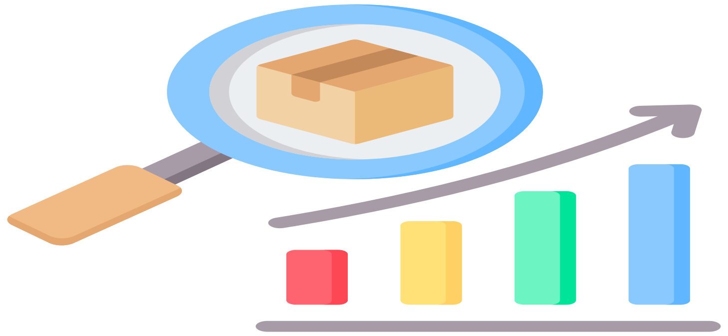
Power BI Examples :

Question: · Which Product Categories generate the highest revenue?

The Health & Beauty category generates the highest revenue at $1.26M, followed closely by Watches & Gifts at $1.21M and Bed, Bath, & Table at $1.04M.

Insights : Lifestyle and personal care products dominate the platform's revenue, suggesting a customer base that prioritizes wellness and gifting .The small revenue gap between the top categories indicates a diversified market where multiple sectors contribute significantly to total growth.





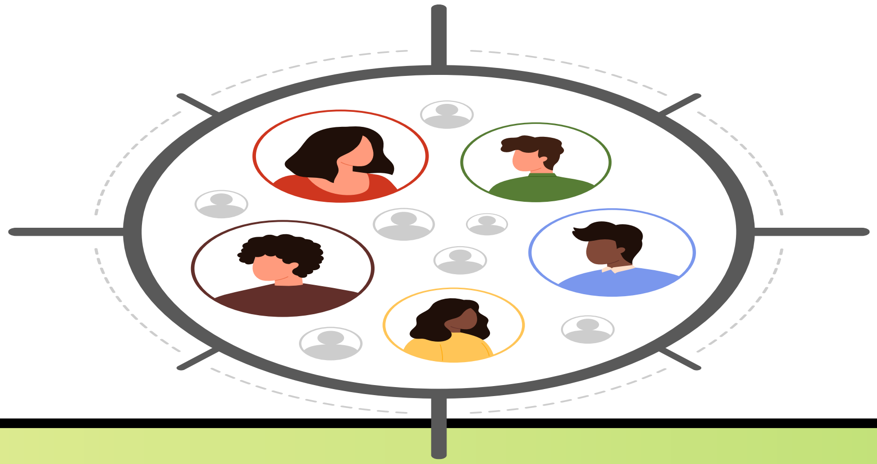
Implementation (Source Code & Execution)

Power BI Examples :

Question: · Which City have the highest number of customers?

Sao Paulo has the highest number of customers by a significant margin, with 15.54K customers.

Insights: Sao Paulo is the primary hub for the business, containing more than double the customers of the second-ranking city, Rio de Janeiro. The concentration of customers in large urban centers like Sao Paulo and Rio de Janeiro suggests that marketing and logistics efforts are most effective in major metropolitan areas.



Implementation (Source Code & Execution)

Power BI Examples :

Question:
- How are Review Scores distributed (1 to 5 stars)?
- What is the Average Review Score by Order Status?

Average Review Score by Status: The "delivered" status holds the highest average review score of 4, while statuses like "shipped," "canceled," and "unavailable" drop to an average of 2, and "processing" has the lowest average at 1.Overall Distribution: The total average review score for all 99,224 reviews is 4, indicating generally high customer satisfaction for completed orders.

Insights: There is a clear correlation between order completion and customer satisfaction, as "delivered" orders significantly outperform all other statuses in review scores. Low scores for "processing" and "shipped" statuses suggest that delays or uncertainties during the fulfillment phase negatively impact customer perception even before delivery.

Review Score Analysis by Order Status

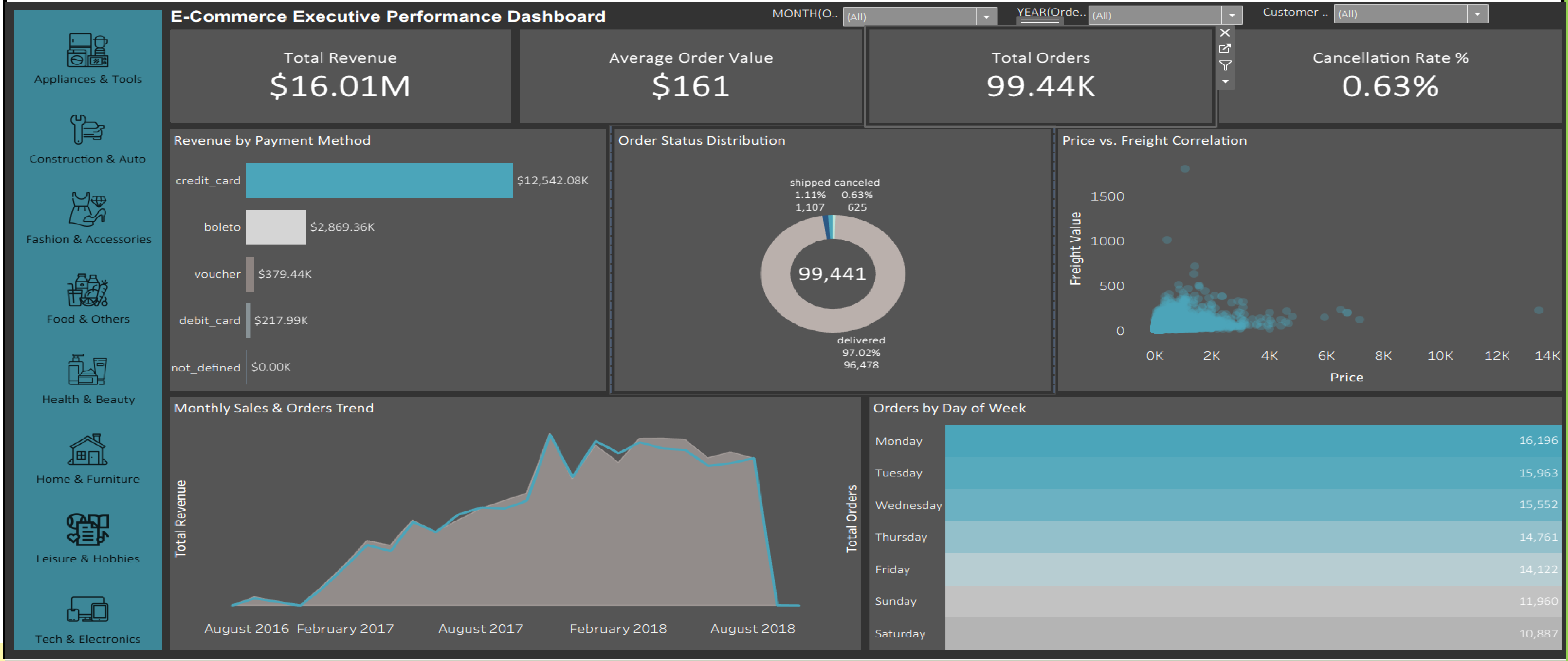
order_status	Total Orders	Count of review_id	Avg Review
delivered	96478	96361	4
shipped	1107	1043	2
canceled	625	609	2
unavailable	609	597	2
invoiced	314	313	2
processing	301	296	1
created	5	3	2
approved	2	2	3
Total	**99441**	**99224**	**4**

Implementation (Source Code & Execution)

5

Track 2 : Tableau Dashboard – Implemented:

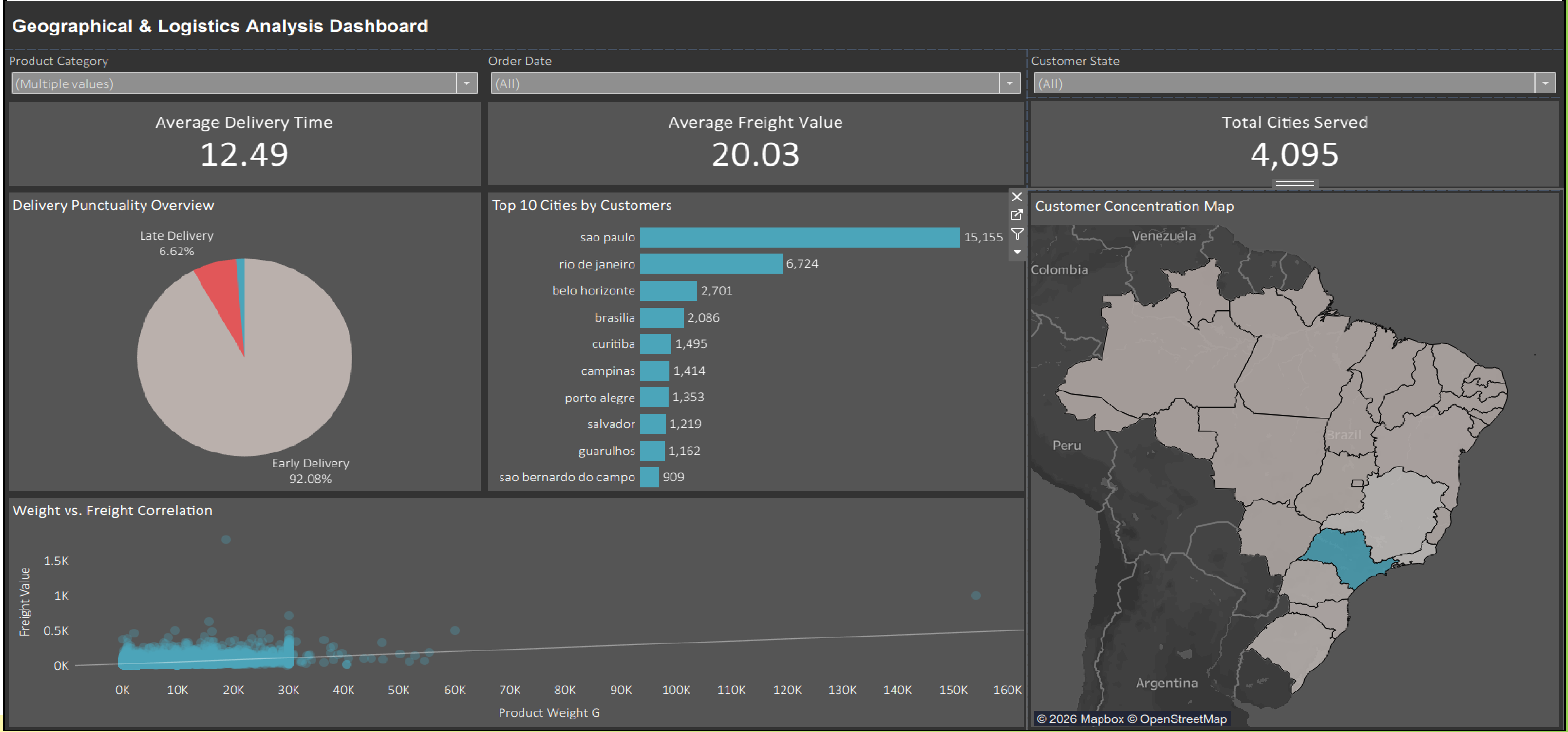
Data Cleaning > Data Analysis Data >Integration >Dashboard Development



Implementation (Source Code & Execution)

5

Tableau Dashboard -



Implementation (Source Code & Execution)

Tableau Examples :

Question: Key Performance Indicators (KPIs)

- What is the Total Revenue generated?
- What is the Total number of unique Orders placed?
- What is the Average Order Value (AOV)?
- What is the Average Delivery Time in days?
- What are the Total Cancelled Orders and the overall Cancellation Rate %?

Total Revenue	Average Order Value	Total Orders	Cancellation Rate %
$16.01M	$161	99.44K	0.63%

Average Delivery Time	Average Freight Value	Total Cities Served
12.49	20.03	4,095

Implementation (Source Code & Execution)

5

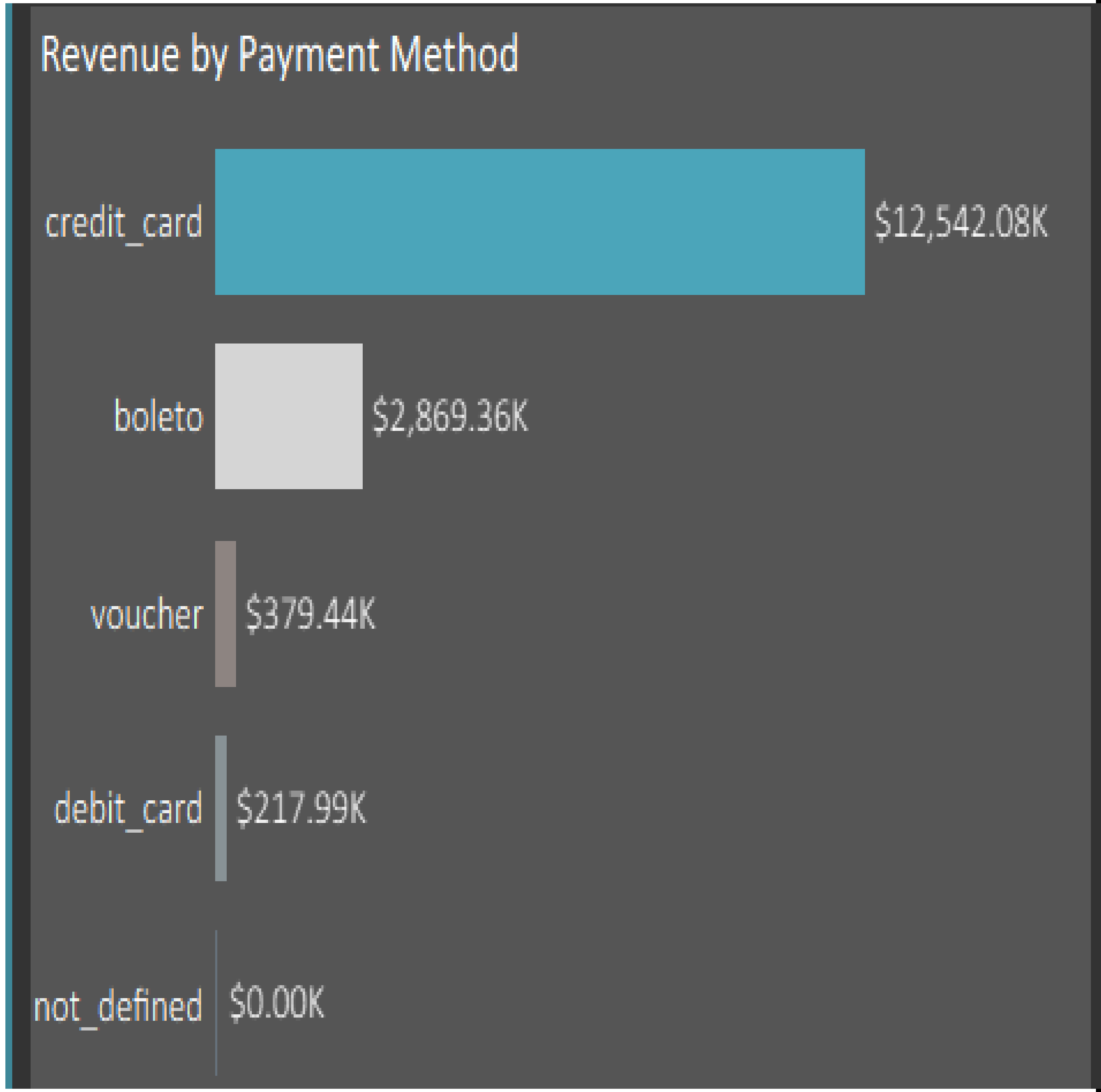
Tableau Examples :

? Question :What is the revenue distribution by Payment Type?

Credit Card payments generated the highest revenue 12,542 k , followed by Boleto, Debit Card, and Voucher payments.

Insights :Customers strongly prefer Credit Card payments, which account for the majority of transactions.

However, over-dependence on one payment method could create future financial or operational risks.



Implementation (Source Code & Execution)

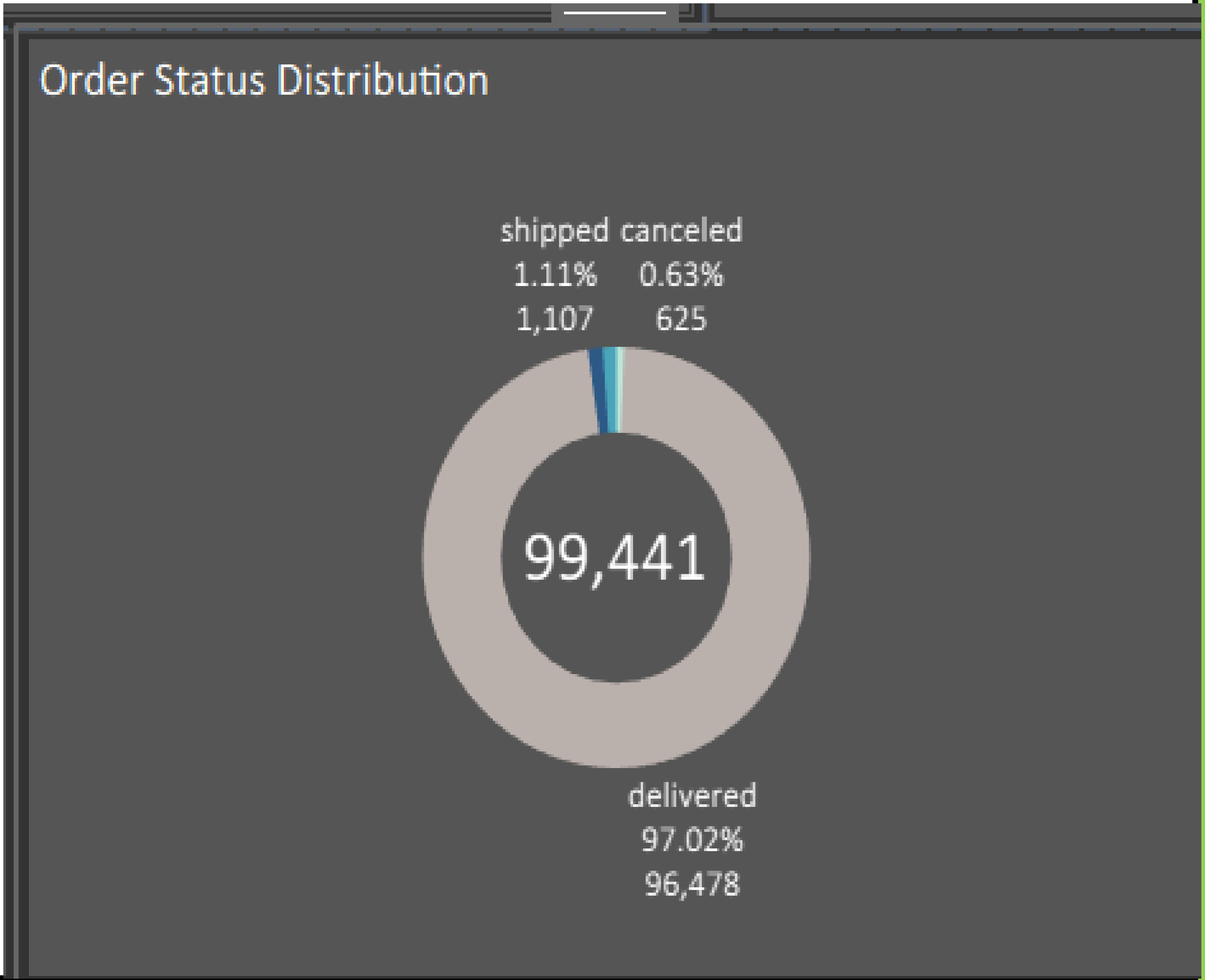
5

Tableau Examples :

Question :What is the distribution of Order Statuses?

Delivered orders represent approximately 97% of total orders, while shipped and canceled orders account for a very small percentage.

Insights :The platform demonstrates high delivery efficiency and a stable fulfillment process with minimal operational failures.



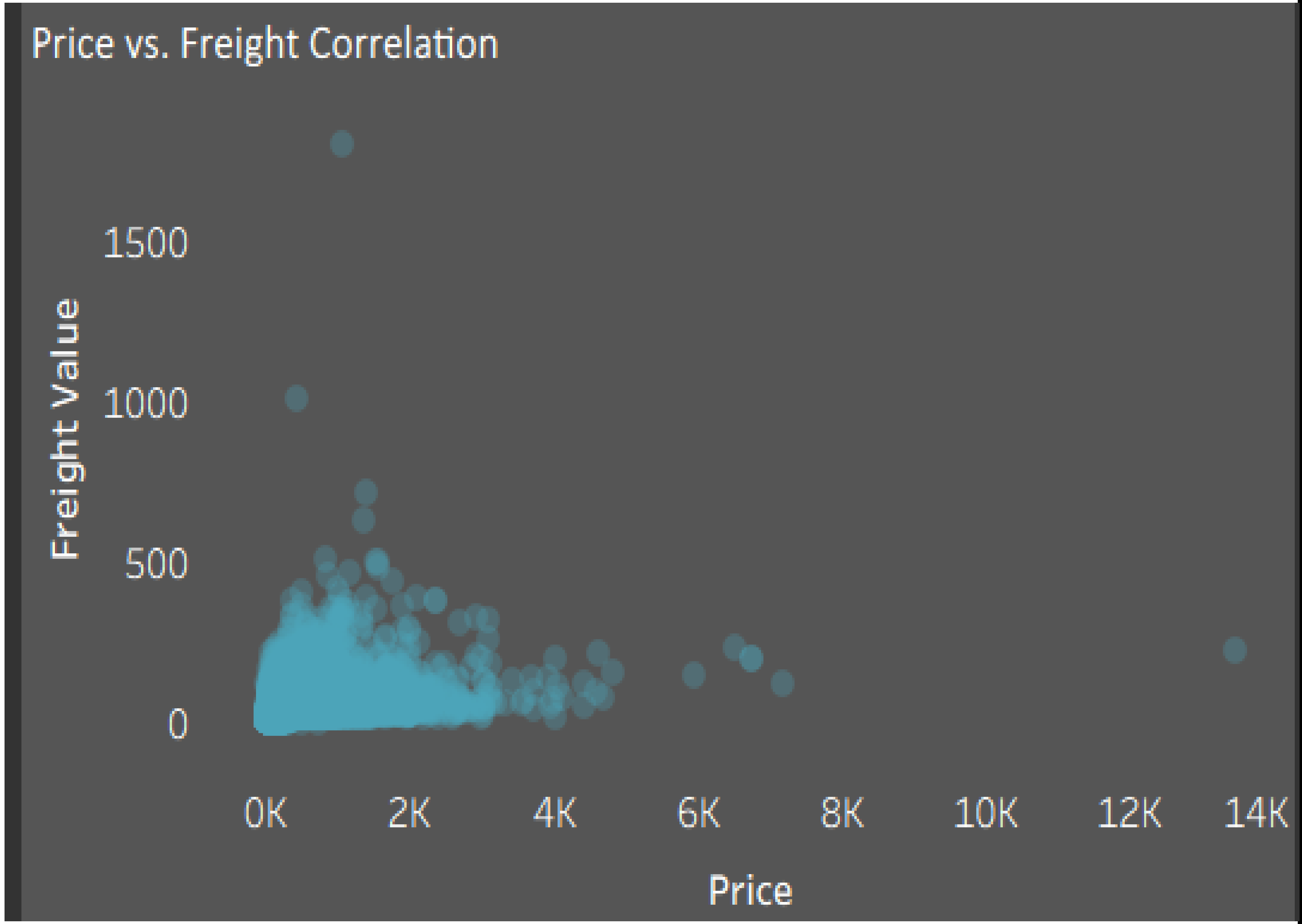
Implementation (Source Code & Execution)

Tableau Examples :

? Question : Is there a correlation between Product Price and Freight Cost?

The scatter plot shows a weak-to-moderate relationship between product price and freight value.

Insights :Shipping cost is influenced more by product weight and delivery distance than by product price itself.



Implementation (Source Code & Execution)

5

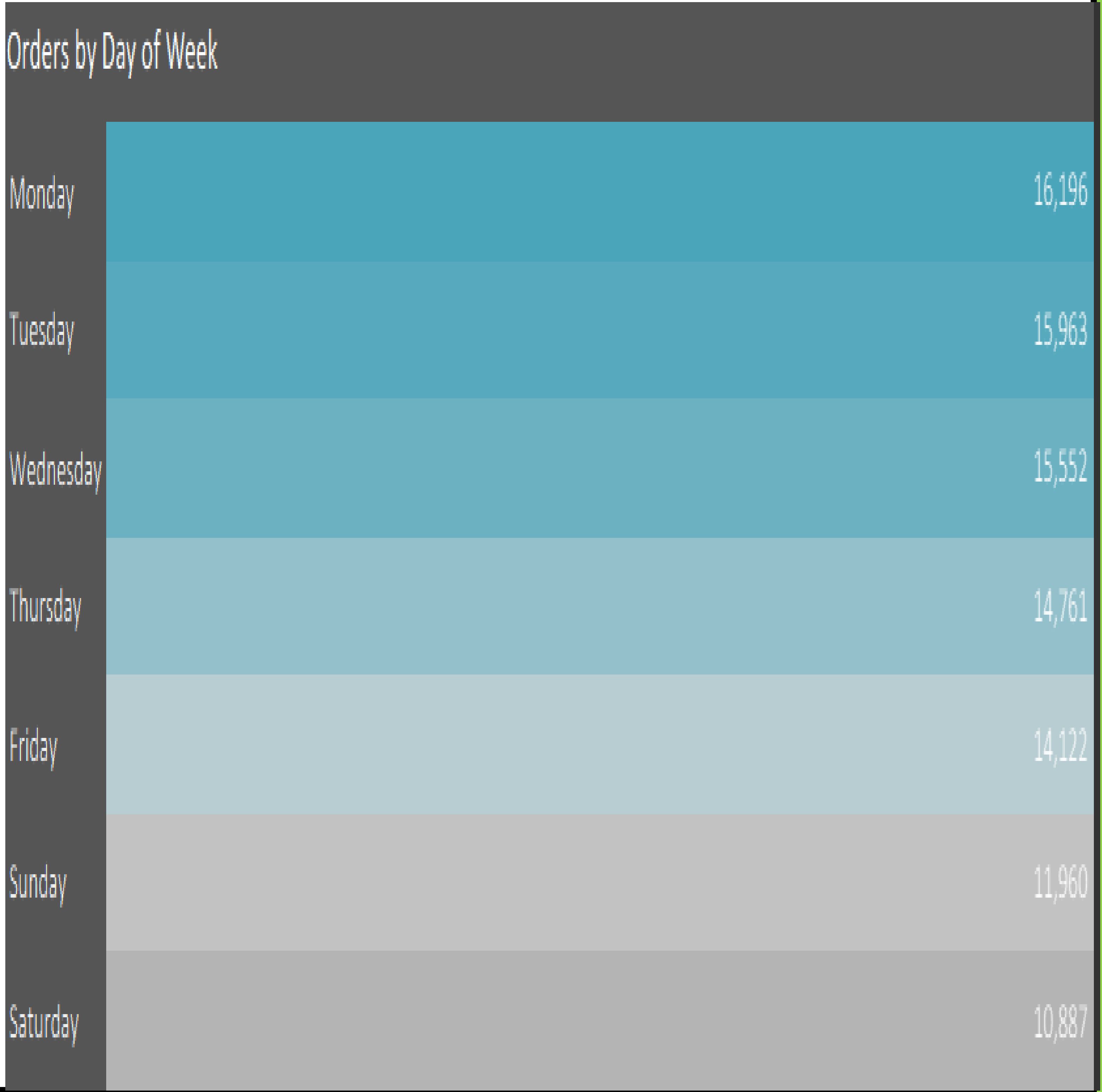
Tableau Examples :

Question: · How are orders distributed by Day of the Week?

Order volume peaks at the beginning of the week, with Monday recording the highest number of orders at 16,196, followed by Tuesday and Wednesday. Activity then gradually declines, reaching its lowest point on Saturday with 10,887 orders.

Insights: The customers prefer shopping at the start of the work week to ensure delivery before the next weekend.

The drop in activity during the weekend (Saturday and Sunday) indicates that promotional campaigns or "flash sales" would be most effective if launched late Sunday or early Monday to capture the peak demand period.



Implementation (Source Code & Execution)

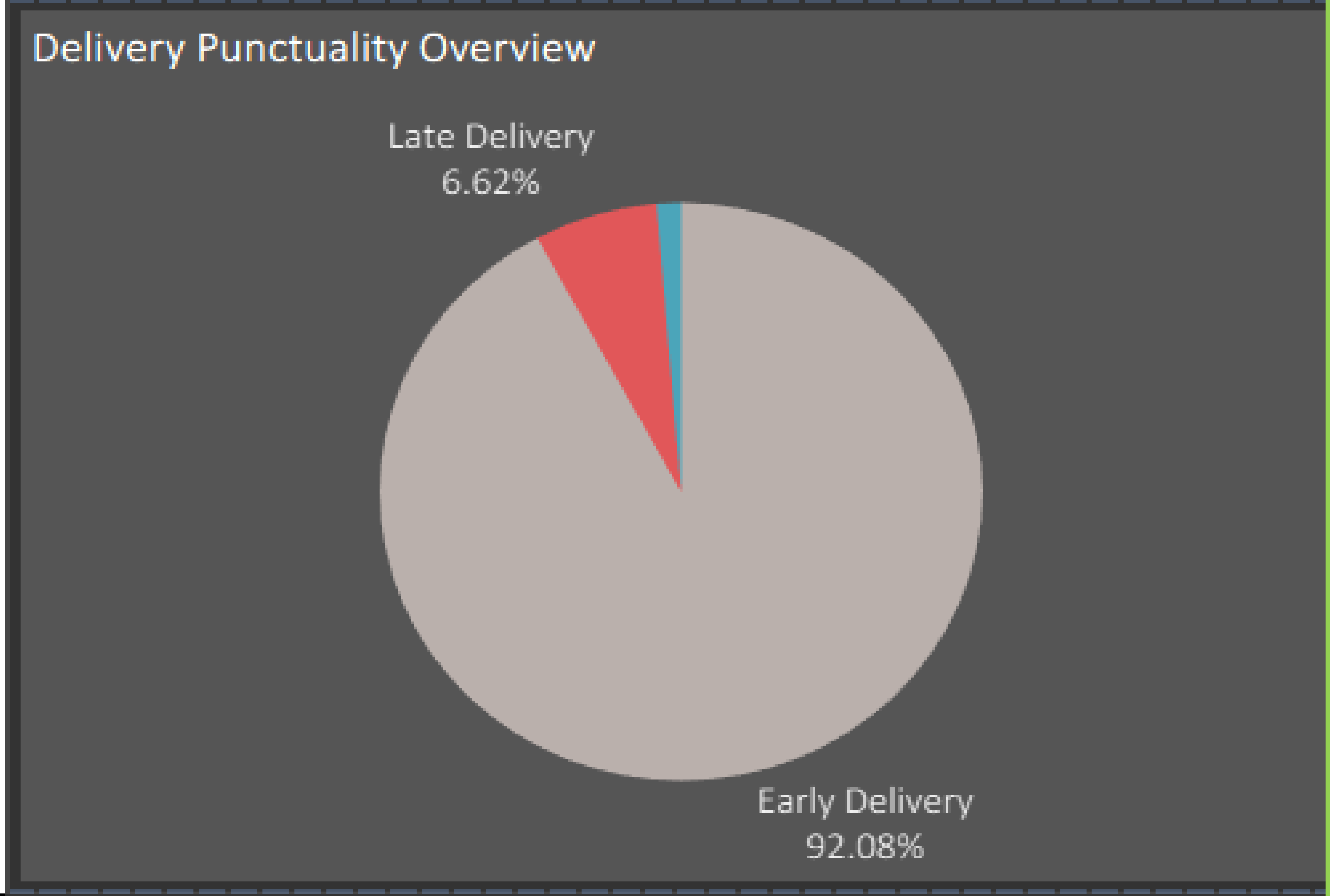
5

Tableau Examples :

❓ Question : What is the Delivery Performance in terms of "On Time" vs. "Late" shipments?

More than 92% of deliveries were completed on time or early, while only around 6.6% experienced delays.

Insights :The logistics system performs efficiently overall, but delayed deliveries significantly affect customer satisfaction and review scores.



Implementation (Source Code & Execution)

5

رواد مصر الرقمية

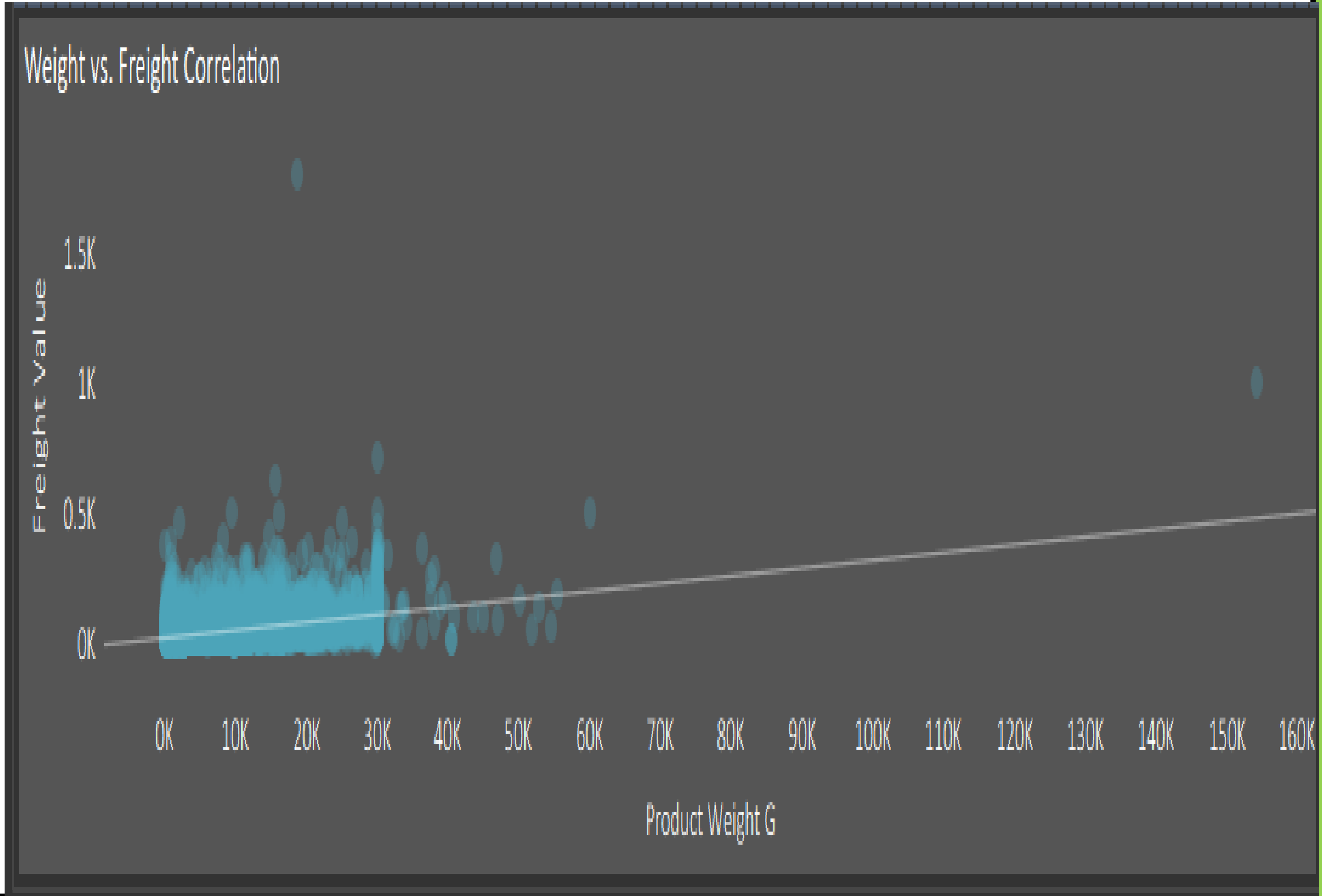
Tableau Examples :

❓ Question : Is there a relationship between Product Weight and Freight Cost?

The scatter plot reveals a positive correlation between product weight and freight value.

Insights :As product weight increases, shipping costs increase significantly.

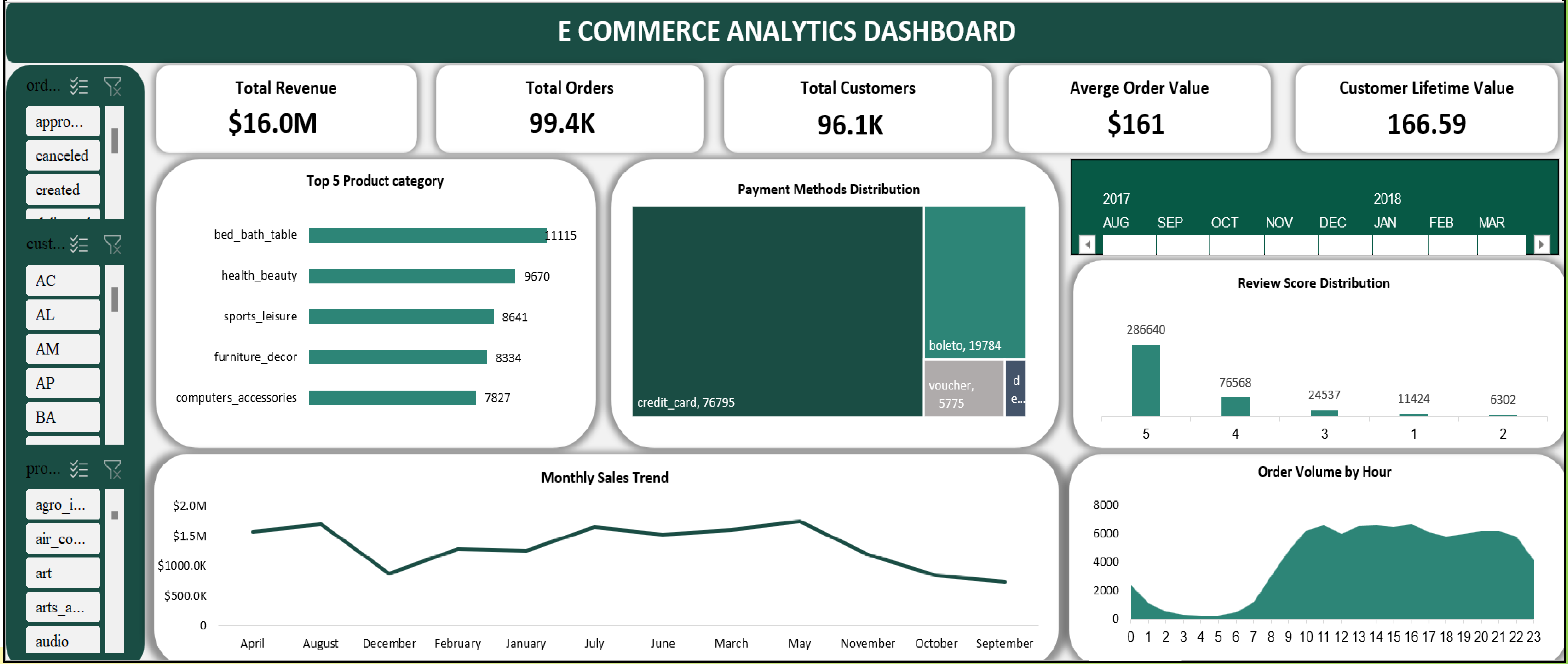
Some products also exhibit unusually high freight costs, indicating potential inefficiencies in logistics pricing or transportation routes..



Implementation (Source Code & Execution)

Track 3 : Excel Dashboard – Implemented:

Data Cleaning > Data Analysis > Data Integration > Dashboard Development



Implementation (Source Code & Execution)

5

Excel Examples :

❓ Question : How are Review Scores distributed?

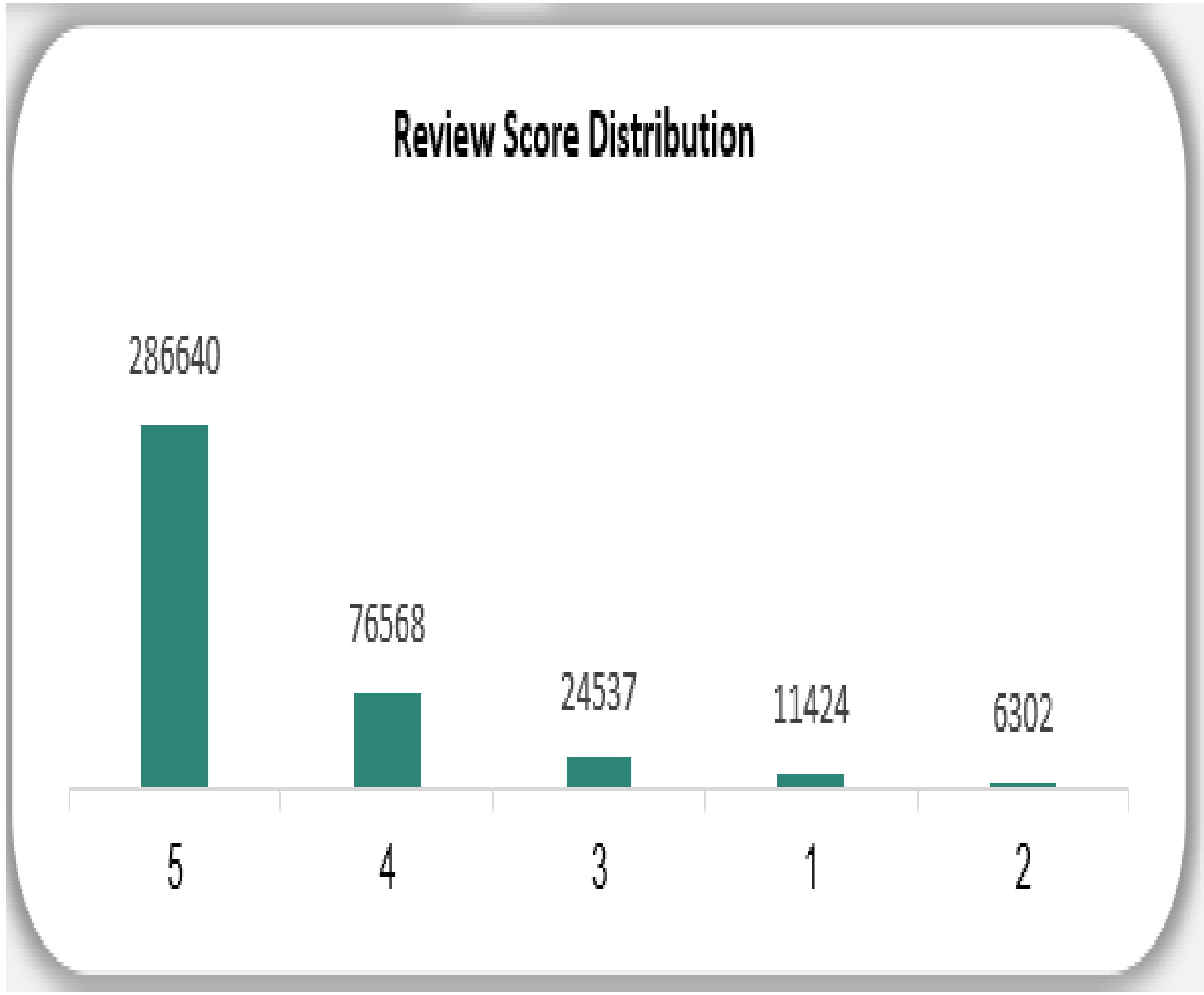
-Most customers gave ratings of:

5 stars

followed by 4 stars

-Low ratings represent a much smaller percentage.

Insights :Customer satisfaction is generally very high, reflecting strong product quality and operational efficiency.



Implementation (Source Code & Execution)

5

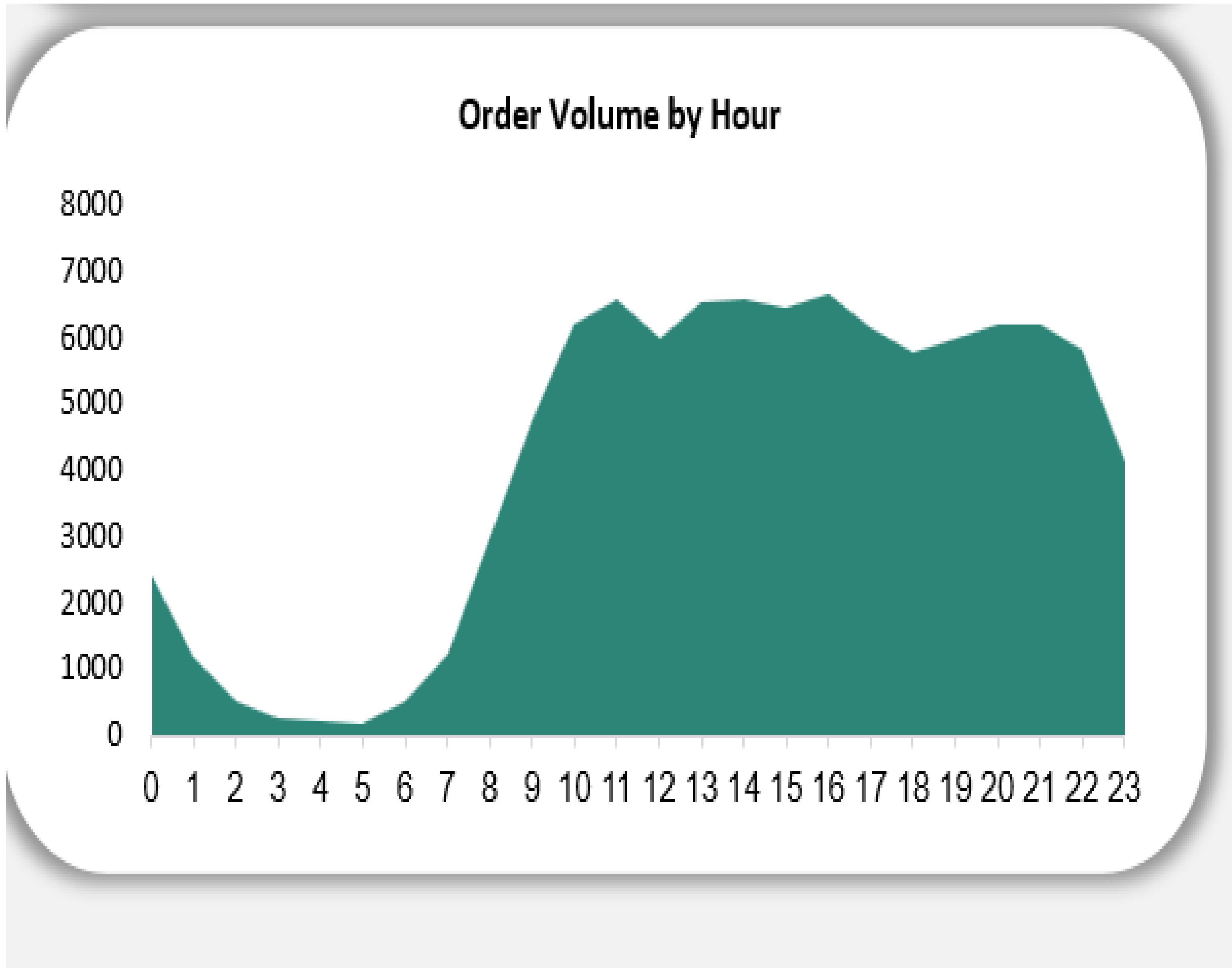
Excel Examples :

❓ Question : What time of day do customers most frequently place orders?

Order activity increases significantly starting from morning hours and peaks between:

10 AM and 10 PM

Lowest activity occurs during late-night and early morning hours.



Insights:

Customers mainly shop during active daytime and evening periods.

This helps optimize:

- marketing campaign timing
- customer support schedules
- and advertising performance.

Final Presentation & Testing& Reports

6

رواد مصر الرقمية

Final Insights Summary -

Market Dominance: Sao Paulo stands as the primary economic hub, contributing significantly to the total customer base of 99,441.

Financial Growth: Revenue achieved an explosive upward trend from 2017 to mid-2018, peaking at $1.19M in November 2017.

Consumer Preference: Credit Cards are the undisputed payment choice (78%), and Health & Beauty is the top-performing category by revenue ($1.26M).

Operational Excellence: The platform maintains an exceptional fulfillment rate with 96,478 successfully delivered orders and an overall average review score of 4/5.

Correlation Between Fulfillment and Rating: The significant gap between the average review score of delivered orders (4/5) versus processing orders (1/5) highlights that customer satisfaction is heavily tied to delivery speed and communication transparency. This suggests that technical or logistical bottlenecks during the initial order phases have a disproportionately negative impact on the brand's reputation, regardless of the final product quality.

Final Presentation & Testing& Reports

6

رواد مصر الرقمية

Recommendations :

Logistics Optimization: Prioritize expanding warehouse capacity and distribution networks in high-density regions like Sao Paulo and Rio de Janeiro to further reduce delivery times.

Payment Strategy: While credit cards lead, enhancing the Boleto and Debit Card user experience could capture underserved market segments that prefer cash-based or immediate payments.

Marketing Timing: Align "Flash Sale" campaigns and promotional emails with Monday and Tuesday mornings to capitalize on the weekly peak in order volume.

Category Expansion: Replicate the successful marketing and supply chain strategies of the Health & Beauty and Watches categories in lower-performing sectors like Auto and Garden Tools to balance revenue streams.

Customer Retention: Implement a proactive feedback loop for "Processing" and "Shipped" statuses to address the lower review scores observed during the fulfillment phase.

Final Presentation & Testing& Reports

6

رواد مصر الرقمية

Recommendations :

Dynamic Pricing & Promotional Strategy: Use the identified peak periods (Mondays and Tuesdays) to implement dynamic pricing or limited-time offers to maximize conversion rates when customer intent is at its highest.

Customer Segmentation & Targeted Marketing: Develop a Loyalty Program specifically for customers in high-density cities like Sao Paulo and Rio de Janeiro to increase "Customer Lifetime Value" (CLV) and reduce acquisition costs.

Basket Analysis & Cross-Selling: Leverage the success of the Health & Beauty and Watches categories by implementing a "Frequently Bought Together" recommendation engine to increase the average order value (AOV).

Boleto Payment Incentives: Offer small "Instant Discounts" for customers using Boleto or Debit Cards to encourage immediate payment and improve the company's cash flow liquidity compared to long-term credit card installments.